Saint Louis University
SCHOOL OF ACCOUNTANCY, MANAGEMENT, COMPUTING AND INFORMATION STUDIES
Department of Information Technology and Computer Science

9349 ITE 17 1:30-3:00 PM M-S

Philippine Educational System — Social Media Discourse Analysis

Aguilan, Jecquar Ravent
Angara, Mark
Barrera, Ryan Adelard
Karim, Syed
Padiernos, Leon Neil
Tibayan, Gelina
Torres, Daniel Linus
Vergara, Marvin John
Wacdagan, Benedict

Submitted to
Mrs. Beverly Ferrer

Date Submitted
June 25, 2026

INTRODUCTION **2**
Team's Journey 2
Background of the Study 3
Objectives of the Study 4
Significance of the Study 5
METHODOLOGY **6**
I. Data Collection 6
II. Dataset Description 6
III. Data Cleaning and Preprocessing 7
IV. Topic Modeling and Evaluation 7
V. Similarity and Dissimilarity Analysis 7
VI. Sentiment Analysis 8
VII. Performance Evaluation 9
VIII. Python Scripts 10
DATA COLLECTION 10
ENVIRONMENT SETUP 14
PHASE 1: DATA EXPLORATION 18
PHASE 1.2: TEXT PREPROCESSING 20
PHASE 2: TOPIC MODELING 31
PHASE 3: PROXIMITY MEASURES 35
PHASE 4: SENTIMENT ANALYSIS 37
PHASE 5: MODEL VALIDATION 49
RESULTS AND DISCUSSION **54**
Frequency Analysis of the Corpus 54
Topic Modeling Results 56
Similarity and Dissimilarity Analysis 58
Sentiment Analysis Results 60
Evaluation of the Topic Model 63
Sentiment Analysis Evaluation 64
CONCLUSION **67**
REFERENCES **68**
APPENDICES **69**

INTRODUCTION

Team's Journey

It was a team effort to conduct this mid-term activity as it required proper planning, distribution of roles, and effective communication within the group. In this case, the team had been assigned the topic "Philippine Educational System," which meant that the team had to perform an analysis of the public discussion regarding the issue, using such methods as web scraping, topic modeling, sentiment analysis, and proximity measures.

As for the flow of the work, the first step of the project was Data Collection, which was done by Aguilan and Tibayan. These students collected data from different social media sites, which include Reddit, TikTok, Facebook, YouTube, and Instagram. To complete this part of the task, they performed web scraping in order to get comments, posts, and discussions related to the Philippine educational system. Once they finished the Data Collection phase, they moved on to Data Exploration and Preprocessing, which is the process of cleaning the text data.

After a successful completion of Data Collection, Model Development was done by Angara, Padiernos, and Torres. It is worth noting that this stage is considered the most challenging since there were three main methods that needed to be implemented: Proximity Measures, where the similarities between comments, posts, and topics have to be found through the use of cosine similarity and Euclidean distance, then, topic modeling, including Latent Dirichlet Allocation (LDA) method for extracting at least three themes related to the discussion; and Sentiment Analysis, where the classification model will be used to find out what kind of sentiments (positive, negative or neutral) the public has towards the mentioned topics.

At the same time, Background and Rationale were the responsibility of Barrera, providing the necessary context and significance for the research. He was also in charge of creating The Team's Journey and Conclusion sections.

The Methodology section was completed by Karim. This stage involves providing a methodological description of all methodologies used in the research, such as data collection, preprocessing procedures, as well as approaches to the calculation of proximity measures, topic modeling, and sentiment analysis. It also included the identification of the most important attributes of the data set.

Lastly, the Discussion of Results section was accomplished by Wacdagan and Vergara. They interpreted the outcomes obtained through topic modeling, sentiment analysis, and proximity measures calculation, which was to result in the interpretation of the obtained results.

From the beginning of the assignment until the end, the group communicated via a group chat in Messenger and the designated lab meetings, where the team discussed the work progress and challenges, and worked together on solving the problems that arose during the implementation of the models. Despite all the restrictions and the difficulties of the project, the team showed resilience and flexibility, which helped make sure that everything went according to the plan.

Background of the Study

The Philippine educational system is an organized structure that facilitates the progression of learners from their basic studies to their chosen professions. There are four major governing bodies in this system: the first being the Department of Education (DepEd), which regulates basic education using the K-12 curriculum, including kindergarten, elementary, junior high, and senior high schools; afterwards, students who finish their studies in the K-12 program have options to enter higher education institutions accredited by the Commission on Higher Education (CHED); learners may also opt to undergo training and acquire technical skills from the Technical Education and Skills Development Authority (TESDA); finally, students who then wish to practice a profession that requires a license are required to take licensure examinations administered by the Professional Regulation Commission (PRC).

The K-12 implementation was designed to conform to international standards by making basic education span twelve years and providing learners with specialized tracks in academics, technical-vocational courses, sports, and the arts. Aside from initiatives by the government to provide free tuition fees in state universities and additional technical-vocational education, several problems persist regarding the system, such as accessibility to education, quality of instruction, learning outcomes, and proper resource allocation (*The Philippine Education System: Complete Guide to CHED, TESDA, DEPED, and PRC*, 2026).

Unfortunately, there are still issues, despite the efforts made by the government to provide education for the future of the Filipino youth. There is a huge crisis in place in that the Philippines has very low scores in terms of creativity and literacy with millions of children not knowing how to read and count properly. It is caused by a number of interrelated problems such as poverty, child labour, early pregnancies, and poor teacher preparation which form an endless loop preventing educational progress. Also, the sector itself does not receive enough funding and many schools have no electricity or clean water. As a result, billions of dollars of money and productivity are being lost yearly due to the unprepared workforce of the country (Denya, 2025).

Likewise, the results obtained from the Second Congressional Commission on Education (EDCOM 2) and PISA 2022 have shown a trend of declining student competence throughout their progression in the educational system. Despite only a third of Grade 3 students showing basic competency in major subjects, this percentage falls down to nearly zero in Senior High School due to a vicious snowball effect caused by the lack of basic competencies taught in primary schools. The Philippines has also been one of the poorest performing countries in the world, with Filipino students achieving scores much lower than the OECD average in terms of reading, mathematics, science, and creativity. Systemic issues identified include blind implementation of the "no student left behind" policy and the pass-through system, which enables students to move forward without having the necessary competencies (Lauengco, 2026). Given that educational issues are continuously drawing attention and raising public concerns, discourse around the state of Philippine education has been increasingly shifting online, where stakeholders, especially young people, can freely express their opinion.

In today's modern era, social media has become a fundamental part of everyday life, especially among the youth. In fact, the Philippines has one of the most digitally-connected populations in the world since it boasts 97.5 million internet users

in January 2025 (Malasig, 2025). Also, according to the National ICT Household Survey, 60% of those between the ages of 10 and 17 years old make use of the internet, and about a third of all Filipino minors have access to it. The level of involvement of digital technology in people's lives has made the social media platforms not only channels of communication but rather public spheres, in which Filipinos voice their opinions and discuss matters of national significance (Abasola, 2025).

In the midst of all of this, the emergence of social media has enabled a digital discussion regarding the worsening education situation in the Philippines. One popular Reddit post which went viral on Facebook and TikTok was posted by a high school teacher, revealing the "nasty truth" about 12 to 13-year-old students who do not even recognize the names of basic shapes and 15 to 18-year-old students who are unable to read and write sentences, spell words, and perform simple addition, subtraction, multiplication, and division. As expected, this caused reactions from netizens and other educators, citing systemic problems such as the "no student left behind" initiative, the fear of failure to receive failing grades by students due to the threat to the rating of the school posed by school principals, and too much bureaucratic paperwork resulting in less time for the production of quality educational materials. This was seen in other online forums as well where teachers complained of being "10 percent teaching and 90 percent administration" and called for a modernized teaching style in accordance with a modern generation (Calinao & Franco, 2024).

This being the case, there is great potential to conduct an analysis of the opinions voiced on the topic of education in the Philippines from social media. Despite the usefulness of statistics that is available in government reports and research papers on the topic, such data fails to capture one important side of the story–the frustration, hopes, and suggestions made by people who have firsthand experience of dealing with this problem. Official statistics give a structured image of the current state of affairs in terms of educational problems, such as literacy rates, enrollment figures, and other indicators; however, this kind of information does not convey the emotions of people involved in the system of education. In this regard, conducting an analysis will allow filling this gap by understanding real opinions of Filipino citizens on the issue.

Objectives of the Study

This study is conducted to offer valuable insights and essential information regarding the public's sentiments towards the Philippine Educational System. By applying specific data mining algorithms to analyze if there are any hidden patterns, this study specifically aims to achieve the following objectives:

1. **To identify major discussion themes** in the social media data using topic modeling, in particular Latent Dirichlet Allocation (LDA), and to find out what topics are the major themes discussed by the public regarding the Philippine educational system.
2. **To measure** the textual similarities using cosine similarity and Euclidean distance among user-generated content regarding the Philippine educational system and to find out which data points have the most and the least similarities.
3. **To evaluate the quality of the topic model** using metrics such as coherence score and perplexity and to determine how coherent the topics generated by the

model are.

4. **To determine** the public's opinions, and sentiments regarding the Philippine educational system across various social media platforms
5. **To develop a rule-based sentiment classifier** utilizing corpus-driven dictionary building to predict the positive, negative, or neutral sentiment of user-generated content and to test the classifier's performance using accuracy, precision, recall, and F1-score via manual validation on a held-out 10% test set.

Significance of the Study

This study aims to contribute to computational social sciences, natural language processing and education policies. The research will facilitate the development of data-driven governance and the practice of mining public opinions, since it will prove how computational methods can be used to obtain perspectives concerning national-level issues. With the growth of influence of social networks on people, it is vital to integrate the technology of text mining in the process of doing research in order to narrow the gap between standard statistical reports and personal experience of citizens. For the policymakers and government agencies, this research will provide an opportunity to use a data-driven method of assessing public sentiments and finding out what the main concerns of Filipino citizens are, ranging from curriculum implementation, teacher welfare, to classroom shortage and literacy rates. The project will help to find at least three major discussion themes and determine if the sentiment towards these topics is positive, negative, or neutral which will enable policymakers to develop responsive policies, allocate budgets, and implement programs to address real needs of students, parents, and teachers. Also, the research will act as a tool for advocacy for teachers, civil society organizations, and other advocacy groups since it will help to voice opinions of people who usually have no chance to express themselves in the policymaking process. Finally, for the academic community and future researchers, this study will illustrate the practical usage of machine learning algorithms such as Latent Dirichlet Allocation (LDA) to perform topic modeling and rule-based classifiers to assess sentiment. Thus, the project will provide a basis for further research in this field by combining publicly available social media data and computational text mining methods.

METHODOLOGY

I. Data Collection

The data used in this study were collected from various social media platforms, namely Reddit, Facebook, TikTok, Youtube, Instagram, and X/Twitter. These platforms were chosen due to their wide range of opinions, experiences, and discussions about social issues, including concerns related to the Philippine educational system. The researchers gathered the data through web scraping using Apify and Python Scripts. Public posts, comments, and discussion threads from 2023 to the present were collected to ensure that the dataset reflected recent educational issues and current public concerns in the Philippines.

To collect relevant discussions, keywords related to education were used, such as Philippine education, K-12, curriculum, DepEd, literacy crisis, and so on. These keywords helped identify discussions relevant to the study. The collected data mainly consist of comments, posts, and public reactions discussing educational quality, government policies, school conditions, teacher-related issues, and student learning outcomes.

II. Dataset Description

The datasets used in this study consisted of textual data collected from social media platforms related to discussions about the Philippine educational system. The raw dataset was collected through web scraping and contained unprocessed discussions, comments, and posts from different platforms. Since the data were directly gathered from social media, the dataset included both relevant and irrelevant records such as duplicate entries, missing values, and noisy data.

After data cleaning and preprocessing, the cleaned and filtered records were compiled into the finalized dataset, which served as the final dataset used for analysis. This dataset contained rows of 971 and columns of 9. Therefore it has more structured and relevant data, making it suitable for similarity analysis, topic modeling, and sentiment analysis.

The following tables present the attributes, data types, and dataset dimensions of the finalized dataset.

Attribute	Data Type	Description
Comment Author	String	Username for commentor
Comment Text	String	Cleaned Comment content
Title	String	Title of post
Post Text	String	Post content
Author	String	Original post author
Date time	Date Time	Date and time posted
Link	String	Source link

Query	String	Search keyword used
Source	String	Social media platform

Table 1. Finalized Dataset Attributes

III. Data Cleaning and Preprocessing

Before conducting the analysis, the collected datasets underwent data cleaning and preprocessing to improve data quality and ensure reliable results. Since social media data are often noisy, unstructured, and inconsistent, preprocessing was necessary to prepare the dataset for similarity analysis, topic modeling, and sentiment analysis. The preprocessing process involved both automated cleaning using Python and manual validation by the researchers.

During preprocessing, duplicate entries, spam comments, incomplete records, and discussions unrelated to the Philippine educational system were identified and removed. The textual data were then standardized by converting all text to lowercase, removing punctuation, special characters, URLs, and emojis are mapped to its meaning to reduce unnecessary noise in the dataset. Then, the text data underwent preprocessing steps such as tokenization, stopword removal, lemmatization, and text vectorization to convert the discussions into a format suitable for topic modeling. Common stopwords such as “the,” “is,” “are,” and “and” were also removed since they contributed confusion to the text analysis. Additionally, some records contained one post with multiple comments grouped together in a single row. To improve analysis accuracy, these grouped comments were separated into individual entries so that each comment could be treated as a separate record. These preprocessing steps helped improve text consistency and ensured that the remaining data focused only on meaningful discussions related to educational issues in the Philippines.

IV. Topic Modeling and Evaluation

Topic modeling was conducted to identify the major themes and recurring issues present in the collected discussions regarding the Philippine educational system. Through topic modeling, the researchers were able to determine the most frequently discussed concerns and issues raised by the public across different social media platforms.

For this study, **Latent Dirichlet Allocation (LDA)** was used as the primary topic modeling technique. LDA is a widely used algorithm for extracting hidden topics from large collections of textual data by identifying patterns in word distribution (Kovelamudi, 2025). Before applying LDA, To evaluate the quality of the generated topics, the researchers used the Coherence Score. This evaluation metric measures how semantically related and meaningful the words within each topic are. By using LDA and coherence score evaluation, the researchers were able to identify and validate the dominant discussion themes related to the Philippine educational system.

V. Similarity and Dissimilarity Analysis

Similarity and dissimilarity analysis was conducted to measure the relationship between discussions in the dataset based on their textual content. This method helped the researchers identify whether comments and posts discussed similar issues or

expressed different opinions regarding the Philippine educational system. Through this analysis, textual discussions were converted into numerical representations, allowing them to be mathematically compared. The analysis focused on identifying similarities among comments related to educational quality, teacher shortages, literacy issues, government policies, and curriculum reforms. Higher similarity scores indicated closely related discussions, while lower similarity scores reflected differing concerns or perspectives.

The study applied four similarity and distance measures, namely **Cosine Similarity, Jaccard Similarity, Euclidean Distance, and Manhattan Distance**. These methods were selected because they are widely used in text mining and data analysis for measuring similarities and differences between datasets. **Euclidean Distance** measures the straight-line distance between data points, while **Manhattan Distance** measures distance based on the sum of absolute differences between variables (Aggarwal, 2023). **Cosine Similarity** measures how closely related discussions are based on the angle between text vectors, making it highly effective for text-based datasets (Jurafsky & Martin, 2025). Meanwhile, **Jaccard Similarity** measures the overlap between sets of words in different comments, helping identify shared terms and common discussion patterns (Aggarwal, 2023). These methods provided multiple perspectives in analyzing similarities and dissimilarities among public discussions regarding the Philippine educational system.

VI. Sentiment Analysis

Sentiment analysis was performed to assess the overall sentiment of the public in relation to the Philippine educational system based on the discussions and comments gathered from social media platforms. This approach enabled researchers to determine the nature of public attitudes toward educational issues, either positive, negative or neutral.

In this study, a rule-based sentiment analysis method using **Corpus-Driven Dictionary Building** was used (Liu, 2024). In this method, each comment or discussion item was categorized as positive, negative, or neutral. Starting from a set of seed words, the lexicon was extended by applying **Pointwise Mutual Information (PMI)**, which analyzes the co-occurrence of words in the collected comments to find semantically related words (Turney, 2022). Positive sentiment was defined as words that indicated approval, improvement, or satisfaction, and negative sentiment as words that indicated criticism, dissatisfaction, or concern. Neutral sentiment was defined as statements that were factual or objective and did not clearly convey positive or negative emotions.

Each comment was examined as a whole sentence rather than as individual words before sentiment classification. The first step involved dividing comments into clauses using contrast connectors such as *pero, but, however,* and *kasi*, giving more weight to the latter clause to capture the final opinion of the comment (Jurafsky & Martin, 2025). Negation handling was applied within a three-token window, allowing phrases such as *“hindi maganda”* to be correctly interpreted as negative instead of positive. This clause-aware approach improved the ability of the system to capture the overall emotional tone of discussions and determine the dominant public sentiment regarding issues in the Philippine educational system (Medhat et al., 2024).

VII. Performance Evaluation

Performance evaluation was conducted to assess the reliability and effectiveness of the methods used in the study. This step was important to determine whether the applied techniques produced meaningful, consistent, and reliable results in analyzing public discussions regarding the Philippine educational system. The evaluation focused on measuring the quality of the similarity analysis, topic modeling, and sentiment analysis results.

For topic modeling, model performance was evaluated using the **Coherence Score**, which measured how meaningful and interpretable the generated topics were. Higher coherence scores indicated better topic quality and stronger relationships between words within each topic (Röder et al., 2025). For similarity and dissimilarity analysis, the results were evaluated by examining how well the applied measures identified related and unrelated discussions within the dataset.

For sentiment analysis, performance was measured through manual inspection of the results produced by the rule-based classifier. The predicted sentiment labels were compared with the interpretations of the comments reviewed by the proponents to assess consistency and correctness (Liu, 2024). This process helped determine whether the sentiment classification accurately reflected the overall public opinion. Through this evaluation process, the researchers ensured that the applied methods produced reliable and meaningful insights from the collected data.

VIII. Python Scripts

DATA COLLECTION

Python Script

```
import os
import re
import time
import requests
import pandas as pd
from apify_client import ApifyClient

# ----- ENVIRONMENT VARIABLES (required) -----
APIFY_TOKEN = os.getenv("APIFY_TOKEN")
if not APIFY_TOKEN:
    raise ValueError("Missing APIFY_TOKEN environment variable")

ACTOR_ID = os.getenv("ACTOR_ID", "RGEBfXu0TSc1siPLb")

SHEET_URL = os.getenv("SHEET_URL")
if not SHEET_URL:
    raise ValueError("Missing SHEET_URL environment variable")

# Optional settings via env
PROXY_COUNTRY = os.getenv("PROXY_COUNTRY", "US")
MAX_RESULTS_PER_KEYWORD = int(os.getenv("MAX_RESULTS_PER_KEYWORD", "5"))
MAX_TOP_LEVEL_COMMENTS = int(os.getenv("MAX_TOP_LEVEL_COMMENTS", "10"))

# ----- SEARCH KEYWORDS (EDUCATION) -----
education_keywords = [
    "Philippine education",
    "DepEd curriculum",
    "K-12",
    "teacher salary",
    "curriculum",
    "tuition fee",
    "senior high school issues",
]
```

Output

Comment_Author	Comment_Text	Title	Post Text	Author	datetime	Link	Query	Social Media Source
paulpjohn	This is more practical than 1k per stude	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
purple_lass	Right? Parang sa Japan at S.Korea, m	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
Hpezlin	This would actually encourage people t	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
wolfram127	Hopefully locally sourced yung mga ing	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
Western-Grocery-6806	Yes, nandyan din yun sa bill na yan. Hi	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
IkigaiSagasu	Yes, children here in PH should be taux	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
angelovepink	Ganda nito. Win-win both educ and agri	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
RecordingBudget2328	Yes please, malaking boost ito sa acad	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
pinkpugita	I'm all for this. Lugaw + egg is very che	Kiko files free breakfast program bill for	"Kiko" Pangilinan on Thursday fc	AshiraLAdonai	2025-07-03 18:1	https://i.redd.it/h	Eudcation Philip	reddit
[deleted]	While the decongestion of the curriculu	DepEd's new curriculum is working for	submitted by /u/Joseph20102011 to r/Philippines	Joseph20102011	2026-01-20 12:0	https://www.phils	Eudcation Philip	reddit
[deleted]	so Sara atleast did something right? If I	DepEd's new curriculum is working for	submitted by /u/Joseph20102011 to r/Philippines	Joseph20102011	2026-01-20 12:0	https://www.phils	Eudcation Philip	reddit
CrispyTomatoFries	I dont super follow events in DepEd pe	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
Setsuna_Lovenotes	walay nasira na yung curriculum kasi]	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
Spelunkie	pero wala pa din may gusto mag-hire d	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
SuperB-3279	well kasalanan na yun ng kung sino ma	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
Un_OwenJoe	Kasi puro pasang awa ayaw ibagsak, p	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
crazyaristocrat66	The policies are there. Sobrang lenient	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
gallifreyfun	May project sila ngayon for struggling r	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
Status_Chance_1526	I guess that makes sense. Pero feels lil	Sagot sa classroom shortage? DepEd	Pinag-aaralan umano ng Department of Educatio	AbanteNewsPH	2025-11-02 2:42	https://i.redd.it/q	Eudcation Philip	reddit
Relevant-Strength-53	The bar is too low sa senado na yung g	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
[deleted]	Educating the younger generations is tl	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
Pinknaballpen	Kudos to Sen. Bam. Yan ang senador.	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
Fragrant_Bid_8123	Tama naman. Bilib ako kay Kiko saka t	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
lesterine817	Ok lahat except #2. Responsibilidad ng	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
Xophosdono	It will help greatly in the meantime sino	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
rlsadiz	True but we're pressed for resources ai	Bam Aquino pushes "whole-of-nation" a	Bam Aquino Pushes "Whole-of-Nation" Approacl	Mindless_Sunda	2025-08-02 8:25	https://i.redd.it/7	Eudcation Philip	reddit
[deleted]	The Philippines was industrializing fron	How good were the Philippine economy	I often see discussions claiming that the Philippin	Sonnybass96	2026-05-01 14:3	https://i.redd.it/je	Eudcation Philip	reddit
Anonymous	- ThisIll wala silang pake, pasa o bags:	To add on with the Education Crisis in t	((PLEASE DONT REPOST/POST ANYWHERE [	Crisis_And_Thrc	2026-03-13 13:0	https://www.redc	Eudcation Philip	reddit
Gone_girl28	Teacher here. As much as only few of t	To add on with the Education Crisis in t	((PLEASE DONT REPOST/POST ANYWHERE [	Crisis_And_Thrc	2026-03-13 13:0	https://www.redc	Eudcation Philip	reddit
BaeTaMi	ThisIl Teacher ang kapatid ko and she	To add on with the Education Crisis in t	((PLEASE DONT REPOST/POST ANYWHERE [	Crisis_And_Thrc	2026-03-13 13:0	https://www.redc	Eudcation Philip	reddit

Remarks

Data Collection In this data collection script, the Apify API scrapes Reddit posts and comments based on specific keywords related to Philippine education. It also organizes the data into a pandas DataFrame, removes duplicates, and exports everything to Google Sheets via a webhook, while also saving a local CSV backup. The output of this automated process is a centralized dataset that includes post titles, raw text, aggregated comment threads, author details, and timestamps. Additionally, it tracks successful and failed logs.

DATA COLLECTION

```
# ----- Build search targets -----
search_targets = []
for kw in education_keywords:
    search_targets.append({
        "query": kw,
        "searchSort": "relevance",
        "timeframe": "year",
        "maxResults":
MAX_RESULTS_PER_KEYWORD,
    })

run_input = {
    "mode": "search",
    "searchTargets": search_targets,
    "includeCommentsMode": "all",
    "maxTopLevelComments":
MAX_TOP_LEVEL_COMMENTS,
    "maxRepliesDepth": 0,
    "includeRaw": False,
    "proxyConfiguration":
{"useApifyProxy": True,
"apifyProxyGroups": ["RESIDENTIAL"]},
    "proxyCountry": PROXY_COUNTRY,
    "tag": "education_search_full",
}

# ----- Initialize Apify client -----
client = ApifyClient(APIFY_TOKEN)
run =
client.actor(ACTOR_ID).call(run_input=r
un_input)

# ----- Download and group -----
all_items =
list(client.dataset(run.default_dataset_i
d).iterate_items())

posts = {}
comments_by_post = {}

for item in all_items:
    if item.get("record_type") == "post":
        permalink = item.get("permalink",
"")
        match =
re.search(r'/comments/([a-zA-Z0-9]+)/',
permalink)
        if not match:
            continue
```

DATA COLLECTION

```
        post_id = match.group(1)
        # Attempt to get the search term
that produced this post
        search_term =
item.get("searchTerm") or
item.get("query") or "Philippine
education"
        posts[post_id] = {
            "title": item.get("title", ""),
            "text": item.get("text", "") or
item.get("selftext", ""),
            "author": item.get("author", ""),
            "source": item.get("source", ""),
            "url": item.get("url") or
permalink,
            "created_utc":
item.get("created_utc_ts", 0),
            "query": search_term,
        }
        comments_by_post[post_id] = []

    elif item.get("record_type") ==
"comment":
        permalink = item.get("permalink",
"")
        match =
re.search(r'/comments/([a-zA-Z0-9]+)/',
permalink)
        if not match:
            continue
        parent_id = match.group(1)
        if parent_id in comments_by_post:

comments_by_post[parent_id].append({
                "body": item.get("text", ""),
                "author": item.get("author",
""),
            })

# ----- Build records: one per post, with
comments combined -----
records = []
for post_id, post_data in posts.items():
    comments =
comments_by_post.get(post_id, [])
    comment_texts = [f"[{c['author']}]
{c['body']}" for c in comments]
    combined_comments =
"\n\n---\n\n".join(comment_texts) if
```

DATA COLLECTION

```
comment_texts else ""

    records.append({
        "Title": post_data["title"],
        "Post Text": post_data["text"],
# column name with space
        "Comments":
combined_comments,                    #
column name exactly
        "Author": post_data["author"],
        "datetime":
pd.to_datetime(post_data["created_utc"]
, unit="s").strftime("%Y-%m-%d
%H:%M:%S") if post_data["created_utc"]
else "",
        "link": post_data["url"],
        "query": post_data["query"],
    })

df = pd.DataFrame(records)
df = df.drop_duplicates(subset=["link"])


# ----- Send to Google Sheets (one row
per post) -----

success_count = 0
fail_count = 0

for idx, row in df.iterrows():
    # Build the payload with exact
column names expected by Apps Script
    item = {
        "Title": row["Title"],
        "Post Text": row["Post Text"],
        "Comments": row["Comments"],
        "Author": row["Author"],
        "datetime": row["datetime"],
        "link": row["link"],
        "query": row["query"],
    }

    try:
        resp = requests.post(SHEET_URL,
json=item, timeout=10)
        if resp.status_code == 200 and
resp.text.strip() == "Success":
            success_count += 1
            if success_count % 10 == 0:
```

DATA COLLECTION		

else:
fail_count += 1
print(f" Row {idx+1} failed: {resp.status_code} {resp.text[:50]}")
except Exception as e:
fail_count += 1
time.sleep(0.1)
 	
if fail_count:
print(f" {fail_count} rows failed.")
 	
# ----- Save local backup ----- 	
df.to_csv("reddit_education_posts_with_comments.csv", index=False)
 	
# ----- Statistics ----- 	
# no comment_count column anymore; we can compute total comments from the combined text 	
total_comments = df["Comments"].str.count(r"---\n\n---").sum() + (df["Comments"] != "").sum() # rough estimate		

ENVIRONMENT SETUP		
Python Script	**Output**	**Remarks**
# Install packages not included by default in Google Colab import subprocess import sys for package in ['gensim', 'wordcloud']: subprocess.check_call([sys.executable, '-m', 'pip', 'install', package, '-q']) print('Colab dependencies installed: gensim, wordcloud')	Colab dependencies installed: gensim, wordcloud	Installs necessary data science libraries (gensim, wordcloud) that are not pre-installed in the Google Colab environment.

ENVIRONMENT SETUP

```
import os
import pandas as pd
import numpy as np
import re
import math
import warnings
from collections import Counter
warnings.filterwarnings('ignore')

# Inline plots for Jupyter / Google Colab
try:

get_ipython().run_line_magic('matplotlib', 'inline')
except Exception:
    pass

import matplotlib.pyplot as plt
import seaborn as sns

try:
    from wordcloud import WordCloud
except ImportError:
    import subprocess, sys

subprocess.check_call([sys.executable, '-m', 'pip', 'install', 'wordcloud', '-q'])
    from wordcloud import WordCloud

import nltk
from nltk.util import ngrams
for _nltk_resource in ['punkt', 'punkt_tab']:
    try:
        nltk.download(_nltk_resource, quiet=True)
    except Exception:
        pass

from nltk.stem import WordNetLemmatizer
import nltk

try:
    from gensim import corpora, models
    from gensim.models import CoherenceModel
except ImportError:
```

```
All libraries loaded successfully.
```

Loaded all necessary libraries for data manipulation, NLP, visualization, and machine learning.

ENVIRONMENT SETUP		
import subprocess, sys subprocess.check_call([sys.executable, '-m', 'pip', 'install', 'gensim', '-q']) from gensim import corpora, models from gensim.models import CoherenceModel from sklearn.feature_extraction.text import TfidfVectorizer from sklearn.decomposition import TruncatedSVD from sklearn.cluster import KMeans from sklearn.metrics.pairwise import cosine_similarity, euclidean_distances, manhattan_distances from sklearn.model_selection import StratifiedKFold, train_test_split from sklearn.metrics import (accuracy_score, precision_score, recall_score, f1_score, confusion_matrix, classification_report) try: plt.style.use('seaborn-v0_8-whitegrid') except OSError: try: plt.style.use('seaborn-whitegrid') except OSError: pass print('All libraries loaded successfully.')		

import os
import pandas as pd
 	
# Works in Google Colab (upload), Google Drive, and local Jupyter 	
CANDIDATE_PATHS = [
'Data Mining Scrape - Fixed Dataset.csv',
'/content/Data Mining Scrape - Fixed Dataset.csv',
'/content/drive/MyDrive/Data Mining Scrape - Fixed Dataset.csv',
		The data loading and cleaning script searches for the scraped dataset in local and Google Drive paths. If the dataset is missing, it prompts for an upload. It then standardizes column names

```
]

DATA_PATH = next((p for p in
CANDIDATE_PATHS if
os.path.exists(p)), None)

if DATA_PATH is None:
    try:
        from google.colab import files
        print('Dataset not found. Please
upload your CSV file:')
        uploaded = files.upload()
        DATA_PATH = next(iter(uploaded))
    except ImportError:
        raise FileNotFoundError(
            'Dataset not found. Place "Data
Mining Scrape - Fixed Dataset.csv" '
            'in the notebook folder or
upload it in Colab.'
        )

print(f'Loading dataset from:
{DATA_PATH}')
df_raw = pd.read_csv(DATA_PATH)


# Select the existing columns with their
correct names from df_raw
df = df_raw[['Comment_Author',
'Comment_Text', 'Title', 'datetime',
'Query','Social Media Source']].copy()


# Rename the columns to the
standardized names for the notebook
df.columns = ['author', 'comment',
'title', 'datetime', 'query', 'source']

# Clean records
df['comment'] =
df['comment'].astype(str).str.strip()
df = df[df['comment'].notna() &
(df['comment'] != '') & (df['comment'] !=
'[deleted]')].copy()
df['source'] =
df['source'].astype(str).str.lower().str.str
ip()
df['source'] = df['source'].replace({'nan':
'unknown'})
df['datetime'] =
pd.to_datetime(df['datetime'],
errors='coerce')
```

	author	comment	title	datetime	query	source	comment_len	word_count
0	pauljpjohn	This is more practical than 1k per student all...	Kiko files free breakfast program bill for K-1...	2025-07-03 18:16:48	Eudcation Philippines	reddit	119	21
1	purple_lass	Right? Parang sa Japan at S.Korea, may free me...	Kiko files free breakfast program bill for K-1...	2025-07-03 18:16:48	Eudcation Philippines	reddit	63	11
2	Hpezlin	This would actually encourage people to go to ...	Kiko files free breakfast program bill for K-1...	2025-07-03 18:16:48	Eudcation Philippines	reddit	53	9

and filters out null, empty, or deleted comments, also calculating basic text metrics like character and word counts. The output shows that the dataset loaded successfully, provides a preview of the first few cleaned rows, and offers initial statistical summaries that reveal average comment lengths across platforms, with Reddit having significantly longer comments than TikTok.Average comment length varies by platform: Reddit (307 chars, 51 words) has the longest comments; TikTok (145 chars, 26 words) has the shortest.

ENVIRONMENT SETUP

```
df['comment_len'] =
df['comment'].str.len()
df['word_count'] =
df['comment'].str.split().str.len()
print(f'Dataset loaded: {len(df)} records')
df.head(3)
```

PHASE 1: DATA EXPLORATION

Python Script	Output	Remarks

`# 1.1 Dataset summary`
`summary = {`
`'Total Records': len(df),`
`'Total Comments': df['comment'].notna().sum(),`
`'Date Range Start': df['datetime'].min(),`
`'Date Range End': df['datetime'].max(),`
`'Exact Duplicate Rows': int(df.duplicated().sum()),`
`'Near-Duplicate Comments (same text)': int(df['comment'].duplicated().sum()),`
`}`
`print('=' * 55)`
`print('DATASET SUMMARY')`
`print('=' * 55)`
`for k, v in summary.items():`
`print(f' {k}: {v}')`
 	
`print('\nPlatform Distribution:')`
`platform_dist = df['source'].value_counts()`
`platform_pct = (platform_dist / len(df) * 100).round(2)`
`platform_table = pd.DataFrame({'Count': platform_dist, 'Percent': platform_pct})`
`print(platform_table.to_string())`
	``
	==========
	DATASET SUMMARY
	==========
	Total Records: 987
	Total Comments: 987
	Date Range Start: 2023-08-14 03:43:22
	Date Range End: 2026-06-17 12:50:32
	Exact Duplicate Rows: 13
	Near-Duplicate Comments (same text): 25
	
	Platform Distribution:
	Count Percent
	source
	facebook 423 42.86
	reddit 290 29.38
	tiktok 230 23.30
	youtube 33 3.34
	instagram 9 0.91
	instagram-hashtag 2 0.20
	``
		Provides a high-level overview of the dataset, including the total number of records, the date range, platform distribution (Facebook, Reddit, TikTok, etc.), and a count of duplicate entries.

`# Deduplicate for modeling while keeping audit trail`	``` Records after comment-level deduplication: 1067 (removed 29) Text Statistics per Platform: comments avg_chars min_chars max_chars avg_words source unknown 1067 195.1 3 4296 32.9 ```	Calculates and displays descriptive statistics for comments, such

PHASE 1: DATA EXPLORATION

```
df_model = df.drop_duplicates(subset=['comment'], keep='first').copy()
print(f'Records after comment-level deduplication: {len(df_model)} (removed {len(df) - len(df_model)})')

# Text statistics per platform
stats = df_model.groupby('source').agg(
    comments=('comment', 'count'),
    avg_chars=('comment_len', 'mean'),
    min_chars=('comment_len', 'min'),
    max_chars=('comment_len', 'max'),
    avg_words=('word_count', 'mean')
).round(1)
print('\nText Statistics per Platform:')
print(stats.to_string())

# Most active threads and contributors
print('\nTop 5 Discussion Threads:')
print(df_model['title'].value_counts().head(5))
print('\nTop 5 Contributors:')
print(df_model['author'].value_counts().head(5))
```

```
Top 5 Discussion Threads:
title
LOOK: The Economy and Devel
Mula sa 20 mag-aaral noong 2
Nagsimula na ang mga school
Sa #ARAL, kasama ang mga gur
TINGNAN: Para sa pagpapatiba
Name: count, dtype: int64

Top 5 Contributors:
author
Anonymous              91
[deleted]               8
notyourhoneysheep       6
Lower-Limit445          4
Kriselda Bartolome      4
Name: count, dtype: int64
```

as character and word counts, broken down by each social media platform. It also lists the top discussion threads and most active contributors.

```
import re
# Language distribution heuristic (Tagalog markers vs English markers)
TAGALOG_MARKERS = {'ang','ng','sa','na','mga','hindi','dapat','guro','paaralan','edukasyon','aral','bata','deped'}
ENGLISH_MARKERS = {'the','and','school','education','teacher','student','because','should','government'}

def detect_language_mix(text):
    toks = set(re.findall(r"[a-zA-Z']+", str(text).lower()))
    tl = len(toks & TAGALOG_MARKERS)
    en = len(toks & ENGLISH_MARKERS)
    if tl >= 2 and en >= 2:
        return 'Taglish'
```

```
Language Distribution:
                 Count  Percent
language_mix
Filipino/Tagalog   544     51.0
English            239     22.4
Mixed/Unclear      188     17.6
Taglish             96      9.0
```

Uses a heuristic to classify each comment's language (e.g., Filipino/Tagalog, English, Taglish) and displays the distribution of these language categories in the dataset.

PHASE 1: DATA EXPLORATION

```
    if tl > en:
        return 'Filipino/Tagalog'
    if en > tl:
        return 'English'
    return 'Mixed/Unclear'

df_model['language_mix'] =
df_model['comment'].apply(detect_lang
uage_mix)
lang_dist =
df_model['language_mix'].value_counts
()
lang_pct = (lang_dist / len(df_model) *
100).round(1)
lang_table = pd.DataFrame({'Count':
lang_dist, 'Percent': lang_pct})
print('Language Distribution:')
print(lang_table.to_string())
```

PHASE 1.2: TEXT PREPROCESSING

Python Script	Output	Remarks

`# Download NLTK data`
`nltk.download('wordnet', quiet=True)`
`nltk.download('omw-1.4', quiet=True)`
 	
`# Initialize lemmatizer`
`lemmatizer = WordNetLemmatizer()`
 	
`all_words = []`
`for c in df_model['comment']:`
 	
`all_words.extend(re.findall(r"[a-zA-Z']+"`
`, str(c).lower()))`
`word_freq = Counter(all_words)`
 	
`AUTO_STOP = {w for w, c in`
`word_freq.most_common(120)`
`if w in {`
 	
`'the','and','for','with','this','that','they','a`
`re','was','were','have','has','had',`
	Custom stopword list size: 92
	Sample before preprocessing:
	This is why I still feel extremely disgusted by the Marcos era. This and the fact that he completely wiped out an entire generation of free-thinking leaders, activists, and youths.
	
	After preprocessing WITH lemmatization:
	['why', 'still', 'feel', 'extremely', 'disgusted', 'marcos', 'era', 'fact', 'completely', 'wiped', 'out', 'entire', 'generation', 'free', 'thinking', 'leader', 'activist', 'youth']
	
	comment / **tokens**
	0 — This is why I still feel extremely disgusted b... — [why, still, feel, extremely, disgusted, marco...
	1 — Marcos ruined everything, he grabbed those cor... — [marcos, ruined, everything, grabbed, those, c...
	2 — That (50-60) decade of progress was easily era... — [decade, progress, easily, erased, succeeding,...
		The preprocessing preparation script initializes NLTK resources and builds a custom bilingual stopword list dynamically by filtering out the most common non-topical words while keeping important educational terms. It also

PHASE 1.2: TEXT PREPROCESSING

'will','would','could','should','not','no','but','so','if','as','by','from','to','of', 'in','on','at','is','it','he','she','we','you','i','a','an','their','them','those', 'these','what','who','how','when','where','why','which','our','more','some','very', 'all','one','about','up','out','time','there','know','think','need','see','look', 'ang','ng','sa','na','ay','at','mga','ko','mo','po','naman','kasi','pero','yung', 'ito','iyon','yon','siya','sila','kami','tayo','natin','nila','niya','nya','nyo', 'ka','ikaw','kayo','ako','pag','kung','kaya','din','rin','lang','lamang','pa', 'nang','dito','doon','saan','sino','ano','bakit','paano','daw','raw','talaga', 'dahil','para','upang','nag','mag','maka','ba','eh','oo','hindi','di','wala','may', 'mayroon','meron','nung','noong','ngayon','lahat','walang','maraming','sobrang', 'medyo','kahit','gaano','ganito','ganyan','ganoon','ganun','siguro','tapos','mas', 'dapat','nga','yan','yun','un','ung','pala','sana','man','wag','huwag','lagi', 'lalo','nasa','dami','puro','nalang','re','like','just','get','got','can','also' }} DOMAIN_KEEP = {'school','schools','teacher','teachers','student','students','education','deped', 'curriculum','grade','college','university','learning','classroom','salary','budget'} CUSTOM_STOPWORDS = AUTO_STOP - DOMAIN_KEEP		defines comprehensive dictionaries for normalizing slang, converting emojis to text, and custom Tagalog lemmatization. The output shows the final size of the custom stopword list and presents a clear before-and-after text sample demonstrating the removal of unnecessary elements. This means that the text cleaning process is localized and context-aware, ensuring that the unique linguistic features, visual sentiment cues, and morphological variations of Filipino social media posts are accurately preserved and standardized for machine learning algorithms.

PHASE 1.2: TEXT PREPROCESSING		

print(f'Custom stopword list size: {len(CUSTOM_STOPWORDS)}')
 	
SLANG_MAP = {
'fckng': 'frustrated', 'amp': 'angry', 'ddshit': 'criticism', 'waley': 'none',
'grabe': 'severe', 'sana': 'hope', 'char': 'joking', 'charot': 'joking'
}		

EMOJI_MAP = { # Positive emojis '😊': 'smiling_face', '😃': 'grinning_face', '😄': 'happy_face', '😁': 'beaming_face', '😆': 'laughing_face', '🥰': 'loving_face', '😍': 'heart_eyes', '🤩': 'star_struck', '😇': 'angel_face', '👍': 'thumbs_up', '👏': 'clapping', '🙌': 'raising_hands', '💪': 'strong', '🎉': 'celebration', '🎊': 'confetti', '✨': 'sparkles', '💯': 'hundred_points', '🔥': 'fire', '❤️': 'red_heart', '💖': 'sparkling_heart', '💕': 'two_hearts', '🙏': 'folded_hands', '👌': 'ok_hand', '🌟': 'glowing_star', '🤗': 'hugging_face', '💐': 'bouquet', '🏆': 'trophy', '🥇': 'first_place', '💡': 'idea', # Negative emojis '😢': 'crying_face', '😭': 'loudly_crying', '😞': 'disappointed', '😔': 'pensive', '😟': 'worried', '😕': 'confused', '🙁': 'slightly_frowning', '😣': 'persevering', '😖': 'confounded', '😫': 'tired_face', '😩': 'weary', '😤': 'huffing', '😠': 'angry_face', '😡': 'pouting', '🤬': 'cursing', '💔': 'broken_heart', '😱': 'screaming', '😨': 'fearful', '😰': 'anxious', '😓': 'downcast', '😥': 'sad_relieved',		EMOJI_MAP Dictionary: Creates a comprehensive mapping of emojis to descriptive text tokens (e.g., '😊': 'smiling_face'), organizing them into positive, negative, and neutral categories for converting visual sentiment cues into machine-readable text. convert_emojis_to_text(text) Function: Replaces all emojis in the input text with their descriptive equivalents and removes any remaining unrecognized

PHASE 1.2: TEXT PREPROCESSING

```
    '👎': 'thumbs_down', '🤢':
'nauseated', '🤮': 'vomiting',
    '💩': 'pile_of_poo', '☠️':
'skull_crossbones', '😈': 'angry_devil',
    '😒': 'unamused', '🙄': 'eye_roll', '😑':
'expressionless','😂': 'tears_of_joy',

    # Neutral / informational
    '🤔': 'thinking', '😐': 'neutral', '😶':
'no_mouth',
    '📚': 'books', '🏫': 'school', '👨‍🏫':
'male_teacher',
    '👩‍🏫': 'female_teacher', '👦': 'boy',
'👧': 'girl',
    '🎓': 'graduation', '📖': 'open_book',
'✏️': 'pencil',
    '🇵🇭': 'philippines', '⚠️': 'warning',
'❗': 'exclamation',
    '❓': 'question', '💬': 'speech', '📢':
'loudspeaker',
}

def convert_emojis_to_text(text):
    """Replace emojis with descriptive
text tokens before cleaning."""
    text = str(text)
    for emoji, name in
EMOJI_MAP.items():
        if emoji in text:
            text = text.replace(emoji, f'
{name} ')
    # Remove any remaining emojis
(those not in map) silently
    text =
re.sub(r'[\U0001F600-\U0001F64F\U
0001F300-\U0001F5FF\U0001F680-
\U0001F6FF'

r'\U0001F1E0-\U0001F1FF\U000027
02-\U000027B0\U0001F900-\U0001
F9FF'

r'\U0001FA00-\U0001FA6F\U0001FA
70-\U0001FAFF\U00002600-\U0000
26FF'

r'\U0000FE00-\U0000FE0F\U00002
00D]+', ' ', text, flags=re.UNICODE)
    return text
```

emojis using regex, ensuring clean text for analysis.

TAGALOG_LEMMA Dictionary: Provides a custom lookup table for Tagalog word normalization, mapping inflected forms of verbs, adjectives, nouns, and adverbs to their base/root forms (e.g., 'nagsabi' → 'sabi', 'maganda' → 'ganda') to standardize variations and improve text analysis accuracy.

PHASE 1.2: TEXT PREPROCESSING

```
# Extended Tagalog lemmatization
TAGALOG_LEMMA = {
    # Verbs
    'nagsabi': 'sabi', 'sinabi': 'sabi',
'sinasabi': 'sabi', 'sasabihin': 'sabi',
    'nagturo': 'turo', 'tinuro': 'turo',
'tinuturo': 'turo', 'itinuturo': 'turo',
    'gumawa': 'gawa', 'ginawa': 'gawa',
'ginagawa': 'gawa', 'gagawin': 'gawa',
    'nagbasa': 'basa', 'binasa': 'basa',
'binabasa': 'basa', 'babasahin': 'basa',
    'nagsulat': 'sulat', 'sinulat': 'sulat',
'sinusulat': 'sulat', 'susulatin': 'sulat',
    'nagbigay': 'bigay', 'ibinigay': 'bigay',
'ibibigay': 'bigay',
    'tumulong': 'tulong', 'tumutulong':
'tulong', 'tutulong': 'tulong',
    'nagbayad': 'bayad', 'binayaran':
'bayad', 'magbabayad': 'bayad',
    'kumain': 'kain', 'kumakain': 'kain',
'kakain': 'kain',
    'umalis': 'alis', 'umaalis': 'alis',
'aalis': 'alis',
    'pumunta': 'punta', 'pupunta':
'punta', 'pumupunta': 'punta',
    'tumanggap': 'tanggap',
'tumatanggap': 'tanggap', 'tatanggap':
'tanggap',

    # Adjectives
    'maganda': 'ganda', 'gumanda':
'ganda',
    'mahirap': 'hirap', 'humihirap':
'hirap', 'nagpapahirap': 'hirap',
    'madali': 'dali',
    'mabigat': 'bigat',
    'malaki': 'laki', 'lumaki': 'laki',
'lumalaki': 'laki',
    'maliit': 'liit', 'lumiliit': 'liit',
    'masaya': 'saya', 'nalulungkot':
'lungkot',
    'malungkot': 'lungkot',

    # Nouns/action words
    'pagtuturo': 'turo', 'pagbabasa':
'basa', 'pagsusulat': 'sulat',
    'pag-aaral': 'aral', 'pagpapaaral':
'aral', 'pagsasanay': 'sanay',
```

PHASE 1.2: TEXT PREPROCESSING		
'pagtulong': 'tulong', 'pagbigay': 'bigay', 'pagbayad': 'bayad', # Common adverbs 'talagang': 'talaga', 'napaka': 'sobra', 'kasing': 'parehas', 'sabay': 'sabay', 'sabay-sabay': 'sabay', }		

def lemmatize_tagalog(token):
"""Lemmatize a Tagalog token using dictionary and pattern matching"""
token_clean = re.sub(r'[^a-z]', '', str(token).lower())
if not token_clean:
return ''
 	
# Direct match 	
if token_clean in TAGALOG_LEMMA:
return TAGALOG_LEMMA[token_clean]
 	
# Check for common Tagalog prefixes 	
for prefix in ['nagpa', 'magpa', 'pinag', 'pina', 'nag', 'mag', 'pag', 'mak', 'nak', 'kin']:
if token_clean.startswith(prefix) and len(token_clean) > len(prefix) + 1:
base = token_clean[len(prefix):]
if len(base) >= 2:
if base in TAGALOG_LEMMA.values():
return base
for suffix in ['an', 'han', 'in', 'ng']:
if base.endswith(suffix) and len(base) > len(suffix) + 1:
stem = base[:-len(suffix)]
if len(stem) >= 2:
return stem
return base
		These three functions work together as a comprehensive text preprocessing pipeline for multilingual Filipino/English text. The lemmatize_tagalog() function handles Tagalog word normalization by first checking a custom dictionary for direct matches, then systematically stripping common prefixes like "nagpa," "magpa," and "nag," followed by suffix removal for patterns like "um," "an," and "han" to extract base word forms.

PHASE 1.2: TEXT PREPROCESSING

```
    # Check for common Tagalog
suffixes
    for suffix in ['um', 'an', 'han', 'in']:
        if token_clean.endswith(suffix)
and len(token_clean) > len(suffix) + 1:
            stem =
token_clean[:-len(suffix)]
            if len(stem) >= 2:
                if stem in
TAGALOG_LEMMA.values():
                    return stem
                return stem

    return token_clean

def lemmatize_token(token):
    """Lemmatize a single token
(English or Tagalog)"""
    token_clean = re.sub(r'[^a-z]', '',
str(token).lower())
    if not token_clean:
        return ''

    # Check if token is likely Tagalog
    tagalog_patterns = ['ng', 'han', 'mag',
'nag', 'pag', 'mak', 'nak']
    is_tagalog =
any(token_clean.startswith(p) or
token_clean.endswith(p) for p in
tagalog_patterns)

    if is_tagalog:
        return
lemmatize_tagalog(token_clean)
    else:
        return
lemmatizer.lemmatize(token_clean)

def
preprocess_with_lemmatization(text,
min_len=3):
    """Full preprocessing pipeline with
lemmatization"""
    text = str(text).lower()

    # Remove URLs
    text = re.sub(r'http\S+|www\S+', ' ',
text)
```

The lemmatize_token() function serves as a dispatcher that identifies whether a token is Tagalog (based on patterns like "ng," "mag," or "nak") or English, applying the appropriate lemmatization method. Finally, preprocess_with_lemmatization() orchestrates the entire pipeline by converting text to lowercase, removing URLs and special characters, tokenizing the text, applying the appropriate lemmatization to each token, and filtering out stopwords and tokens shorter than a specified minimum length (default 3 characters), producing a clean list of normalized tokens ready for downstream NLP

PHASE 1.2: TEXT PREPROCESSING		

# Remove special characters (keep letters and spaces) 	
text = re.sub(r'[^a-z\s]', ' ', text)
 	
# Tokenize and lemmatize 	
tokens = text.split()
lemmatized = [lemmatize_token(t) for t in tokens]
 	
# Remove stopwords and short tokens 	
final_tokens = [t for t in lemmatized
if t
and t not in CUSTOM_STOPWORDS
and len(t) >= min_len
and t.isalpha()]
 	
return final_tokens
		tasks like topic modeling and sentiment analysis.

df_model['tokens'] = df_model['post_text'].apply(preprocess_with_lemmatization) # Changed from 'comment' to 'post_text'
 	
print("Sample before preprocessing:")
print(df_model['post_text'].iloc[0]) # Changed from 'comment' to 'post_text'
print("\nAfter preprocessing WITH lemmatization:")
print(df_model['tokens'].iloc[0])
 	
def preprocess_for_sentiment(text):
t = str(text).lower()
# 1. Convert emojis to text FIRST (before slang replacement) 	
t = convert_emojis_to_text(t)
# 2. Apply slang normalization 	
for slang, replacement in SLANG_MAP.items():
t = t.replace(slang, f {replacement} ')
return t
 	
# Apply sentiment preprocessing to a separate column if needed 			

PHASE 1.2: TEXT PREPROCESSING

```
df_model['sentiment_text'] =
df_model['comment'].apply(preprocess_
for_sentiment)


df_model[['comment', 'tokens']].head(3)
```

```
# 1.3 Frequency analysis — overall
and per platform
all_tokens = [t for toks in
df_model['tokens'] for t in toks]
top20_words =
pd.DataFrame(Counter(all_tokens).mo
st_common(20), columns=['Word',
'Frequency'])

all_bigrams = [bg for toks in
df_model['tokens'] for bg in
ngrams(toks, 2)]
top20_bi = pd.DataFrame([(' '.join(k), v)
for k, v in
Counter(all_bigrams).most_common(2
0)], columns=['Bigram', 'Frequency'])

all_tri = [tg for toks in
df_model['tokens'] for tg in
ngrams(toks, 3)]
top20_tri = pd.DataFrame([(' '.join(k),
v) for k, v in
Counter(all_tri).most_common(20)],
columns=['Trigram', 'Frequency'])

print('Top 20 Words:')
print(top20_words.to_string(index=Fal
se))
print('\nTop 20 Bigrams:')
print(top20_bi.to_string(index=False))
print('\nTop 20 Trigrams:')
print(top20_tri.to_string(index=False))

# Per-platform top words table
platform_top = {}
for src in df_model['source'].unique():
    toks = [t for toks in
df_model[df_model['source']==src]['toke
ns'] for t in toks]
    platform_top[src] =
Counter(toks).most_common(10)
```

```
Top 20 Words:
      Word  Frequency
    school        219
   teacher        216
   student        168
 education        150
     deped        112
curriculum        107
    system         94
   subject         87
   college         71
     grade         71
      bata         62
      year         61
     honor         56
      even         53
   matatag         53
       yes         50
      need         48
      guro         48
philippine         46
     class         45
```

Computes and displays the most frequent words, bigrams, and trigrams across the entire dataset, as well as the top words per platform. These results are used to understand the dominant vocabulary of the discourse.

PHASE 1.2: TEXT PREPROCESSING

```
print('\nPer-platform top words computed for', len(platform_top), 'sources')
```

```
Top 20 Bigrams:
              Bigram  Frequency
matatag curriculum          45
  education system          33
       high school          22
     public school          20
   salary increase          17
       senior high          17
    private school          13
         araw araw          11
         sir jeric          11
   brigada eskwela          10
   basic education          10
   feeding program          10
         sec sonny           9
   student teacher           8
        sec angara           8
 critical thinking           8
         god bless           8
 learning material           8
 quality education           7
     social medium           7

Top 20 Trigrams:
                                 Trigram  Frequency
               department education deped        153
basahin depedphilippines bagongpilipinas        100
                  secretary sonny angara         89
                education secretary sonny         77
                       school year basahin         49
           year basahin depedphilippines          49
         second congressional commission          48
       congressional commission education          48
                   term school calendar           47
                      three term school           46
                  grading period system           46
               implementation three term           43
                    school based feeding           43
                   based feeding program           43
                 non teaching personnel            42
                         deped land bank           42
                  land bank philippine             42
                    mula sariling bulsa            41
                 sariling bulsa hahanda            41
           academic recovery accessible            40
```

PHASE 1.2: TEXT PREPROCESSING

```
# Phase 1 visualizations
fig, axes = plt.subplots(2, 2,
figsize=(14, 10))

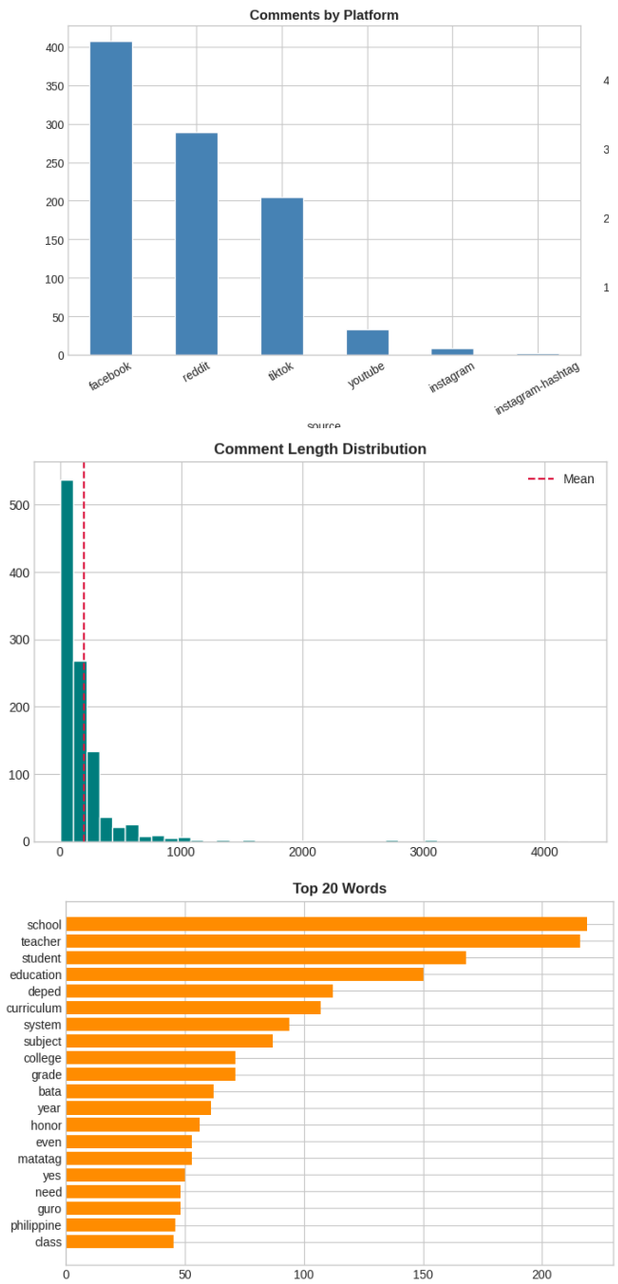
platform_dist_model =
df_model['source'].value_counts()
platform_dist_model.plot(kind='bar',
ax=axes[0,0], color='steelblue',
edgecolor='white')
axes[0,0].set_title('Comments by
Platform', fontweight='bold')
axes[0,0].tick_params(axis='x',
rotation=30)

axes[0,1].hist(df_model['comment_len'],
bins=40, color='teal', edgecolor='white')
axes[0,1].axvline(df_model['comment_l
en'].mean(), color='crimson',
linestyle='--', label='Mean')
axes[0,1].set_title('Comment Length
Distribution', fontweight='bold')
axes[0,1].legend()

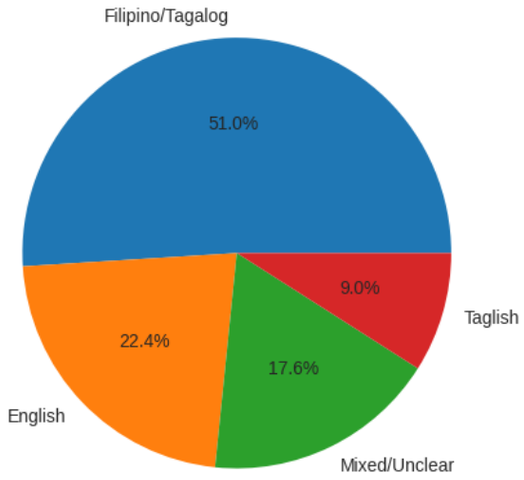
axes[1,0].barh(top20_words['Word'][::-1
], top20_words['Frequency'][::-1],
color='darkorange')
axes[1,0].set_title('Top 20 Words',
fontweight='bold')

lang_dist.plot(kind='pie',
autopct='%1.1f%%', ax=axes[1,1])
axes[1,1].set_title('Language Mix',
fontweight='bold')
axes[1,1].set_ylabel('')

plt.tight_layout()
plt.savefig('phase1_exploration.png',
dpi=150)
plt.show()
```



Generates visualizations to explore the data, including a bar chart for platform distribution, a histogram for comment length, a horizontal bar chart for top words, and a pie chart for language mix.

PHASE 1.2: TEXT PREPROCESSING		
		

PHASE 2: TOPIC MODELING		
PYTHON SCRIPT	**OUTPUT**	**REMARKS**

tokenized_docs = [t for t in df_model['tokens'].tolist() if len(t) >= 2]
dictionary = corpora.Dictionary(tokenized_docs)
dictionary.filter_extremes(no_below=2, no_above=0.95) # Lowered no_below from 3 to 2
dictionary.compactify()
corpus_bow = [dictionary.doc2bow(t) for t in tokenized_docs]
 	
tfidf_vectorizer = TfidfVectorizer(
max_features=5000,
 	
stop_words=list(CUSTOM_STOPWORDS),
min_df=2, max_df=0.95, # Lowered min_df from 5 to 2
ngram_range=(1, 3)
)
doc_strings = [' '.join(t) for t in tokenized_docs]
tfidf_matrix = tfidf_vectorizer.fit_transform(doc_strings)
feature_names = tfidf_vectorizer.get_feature_names_out(
	Corpus size: 945 | Vocab: 2980 | TF-IDF: (945, 5000)	The topic modeling preparation script changes the preprocessed tokenized documents into a Bag-of-Words (BOW) corpus using Gensim's Dictionary. It filters out extreme frequencies and generates a TF-IDF matrix with Scikit-learn's TfidfVectorizer, including custom stopwords and This means the textual data has been converted into numerical formats needed by machine learning

PHASE 2: TOPIC MODELING

) print('Corpus size:', len(corpus_bow), '| Vocab:', len(dictionary), '| TF-IDF:', tfidf_matrix.shape)		algorithms. It establishes the foundational space where the relationships between words and documents can be measured and analyzed by the LDA model.

# LDA evaluation 	
lda_results = {}
print(f"{'Topics':>8} {'Coherence':>12} {'Perplexity':>12}")
print('-' * 38)
for n in [3, 5, 7, 10]:
  lda_m = models.LdaModel(corpus=corpus_bow, id2word=dictionary, num_topics=n, random_state=42, passes=20, alpha='auto')
  coh = CoherenceModel(model=lda_m, texts=tokenized_docs, dictionary=dictionary, coherence='c_v', processes=1).get_coherence()
  per = lda_m.log_perplexity(corpus_bow)
  lda_results[n] = {'coherence': coh, 'perplexity': per, 'model': lda_m}
  print(f'{n:>8} {coh:>12.4f} {per:>12.4f}')
 	
best_lda_n = max(lda_results, key=lambda x: lda_results[x]['coherence'])
	Topics Coherence Perplexity
	3 0.2936 -6.9821
	5 0.4924 -6.7876
	7 0.4540 -6.6896
	10 0.4895 -6.5741
		The LDA evaluation script trains multiple Latent Dirichlet Allocation models ranging from 3 to 10 topics. It calculates the coherence score and perplexity for each model to find the optimal number of topics and extracts top keywords to give readable names to the final selected model. The output displays a comparison table of the evaluation metrics, confirms the choice of 5 topics based on the best balance of coherence and interpretability, and lists the topic names.

PHASE 2: TOPIC MODELING

```
# Select optimal topic count using combined criteria
OPTIMAL_N = 5 # balance coherence, interpretability, distribution
best_lda = models.LdaModel(corpus=corpus_bow, id2word=dictionary,
num_topics=OPTIMAL_N,
                           random_state=42,
eval_every=10, passes=20,
alpha='auto')

def lda_top_words(model, topic_id, n=10):
    return [dictionary[id_] for id_, _ in model.get_topic_terms(topic_id, topn=n)]

topic_keywords = {i: lda_top_words(best_lda, i) for i in range(OPTIMAL_N)}

# Topic names derived ONLY from dataset keywords
def make_topic_name(keywords):
    return ' & '.join([k.title() for k in keywords[:3]])

TOPIC_NAMES = {i: make_topic_name(topic_keywords[i]) for i in range(OPTIMAL_N)}

print('LDA Topics (dataset-derived names):')
for i in range(OPTIMAL_N):
    print(f'  T{i}: {TOPIC_NAMES[i]} | Keywords: {', '.join(topic_keywords[i][:8])}')
```

```
LDA Topics (dataset-derived names):
  T0: Curriculum & Deped & Student | Keywords: curriculum, deped, student, matatag, program, philippine, subject,
  T1: Learning & System & Commission | Keywords: learning, system, commission, philippine, student, development, y
  T2: Aral & Learning & Guro | Keywords: aral, learning, guro, deped, program, tutor, act, ito
  T3: Deped & Department & Student | Keywords: deped, department, student, bagongpilipinas, bansa, depedphilippine
  T4: Loan & Deped & Financial | Keywords: loan, deped, financial, teaching, personnel, depedphilippines, bagongpi
```

Topics align with educational discourse: Curriculum & DepEd, Learning & System, Aral & Guro, Department & Student, and Loan & Financial.

PHASE 2: TOPIC MODELING

```
def assign_topic(doc_bow):
    topics =
best_lda.get_document_topics(doc_bo
w)
    return max(topics, key=lambda x:
x[1])[0] if topics else 0

df_valid =
df_model.iloc[:len(corpus_bow)].copy()
df_valid['topic_id'] = [assign_topic(d) for
d in corpus_bow]
df_valid['topic_name'] =
df_valid['topic_id'].map(TOPIC_NAMES)

topic_counts =
df_valid['topic_id'].value_counts().sort_i
ndex()
total_docs = len(df_valid)

topic_table = pd.DataFrame({
    'Topic ID': list(range(OPTIMAL_N)),
    'Topic Name': [TOPIC_NAMES[i] for i
in range(OPTIMAL_N)],
    'Top Keywords': ['
'.join(topic_keywords[i][:6]) for i in
range(OPTIMAL_N)],
    'Comment Count':
[topic_counts.get(i, 0) for i in
range(OPTIMAL_N)],
    'Percentage':
[f'{topic_counts.get(i,0)/total_docs*100
:.1f}%" for i in range(OPTIMAL_N)],
    'Per-Platform': [''] * OPTIMAL_N,
})

# Per-platform distribution for top
topics
for i in range(OPTIMAL_N):
    plat =
df_valid[df_valid['topic_id']==i]['source']
.value_counts().to_dict()
    topic_table.loc[topic_table['Topic ID']
== i, 'Per-Platform'] = str(plat)

print('Top Topics:')
topic_table
```

Top Topics:

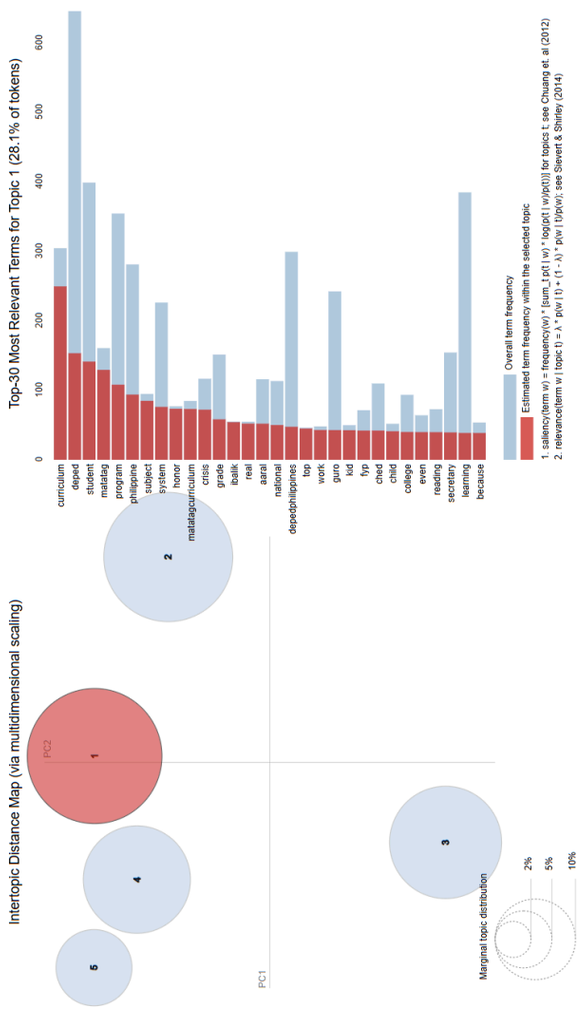
	Topic ID	Topic Name	Top Keywords	Comment Count	Percentage	Per-Platform
0	0	Curriculum & Deped & Student	curriculum, deped, student, matatag, program, ...	455	48.1%	{'reddit': 239, 'tiktok': 134, 'facebook': 49,...
1	1	Learning & System & Commission	learning, system, commission, philippine, stud...	154	16.3%	{'facebook': 124, 'reddit': 17, 'instagram': 6...
2	2	Aral & Learning & Guro	aral, learning, guro, deped, program, tutor	70	7.4%	{'facebook': 68, 'reddit': 1, 'tiktok': 1}
3	3	Deped & Department & Student	deped, department, student, bagongpilipinas, b...	169	17.9%	{'facebook': 95, 'tiktok': 48, 'reddit': 26}
4	4	Loan & Deped & Financial	loan, deped, financial, teaching, personnel, d...	97	10.3%	{'facebook': 71, 'tiktok': 20, 'reddit': 6}

The topic assignment script uses the optimized 5-topic LDA model to determine the main topic for each document. It then aggregates this information to create a summary table showing the comment count, percentage distribution, and breakdown by platform for each topic. The output is a detailed report showing that topics like "Curriculum & DepEd & Student" dominate the discussion (e.g., 48.1%), while financial or teacher -focused topics have smaller shares, along with their specific platform distributions.

```
import subprocess, sys
subprocess.check_call([sys.executable,
'-m', 'pip', 'install', 'pyLDAvis', '-q'])
```

```
pyLDAvis installed.
```

Installs the pyLDAvis library required

PHASE 2: TOPIC MODELING		
print('pyLDAvis installed.')		to generate the interactive intertopic distance visualization from the trained LDA model.
import pyLDAvis import pyLDAvis.gensim_models as gensimvis pyLDAvis.enable_notebook() lda_vis = gensimvis.prepare(best_lda, corpus_bow, dictionary, sort_topics=False) pyLDAvis.save_html(lda_vis, 'lda_vis_interactive.html') print('Saved → lda_vis_interactive.html') lda_vis # renders inline in Colab		Prepares and renders an interactive pyLDAvis panel using the optimized 5-topic LDA model, displaying the intertopic distance map via multidimensional scaling and the top-30 most relevant terms per topic weighted by λ. The visualization is also exported as a standalone HTML file for offline viewing.

PHASE 3: PROXIMITY MEASURES		
Python Script	**Output**	**Remarks**
# Topic-level TF-IDF documents		Computed

PHASE 3: PROXIMITY MEASURES

```
topic_texts = []
for t in range(OPTIMAL_N):
    subset = df_valid[df_valid['topic_id'] == t]['comment'].astype(str).tolist()
    topic_texts.append(' '.join([' '.join(preprocess_with_lemmatization(c)) for c in subset]))

prox_vec = TfidfVectorizer(max_features=500, stop_words=list(CUSTOM_STOPWORDS), min_df=1, ngram_range=(1,2))
topic_vecs = prox_vec.fit_transform(topic_texts)
prox_feats = prox_vec.get_feature_names_out()

cos_sim = cosine_similarity(topic_vecs)
euc_dist = euclidean_distances(topic_vecs.toarray())
man_dist = manhattan_distances(topic_vecs.toarray())

# Jaccard on top keywords per topic
def top_keyword_set(tid, k=20):
    row = topic_vecs[tid].toarray().flatten()
    idx = row.argsort()[-k:][::-1]
    return set(prox_feats[i] for i in idx)

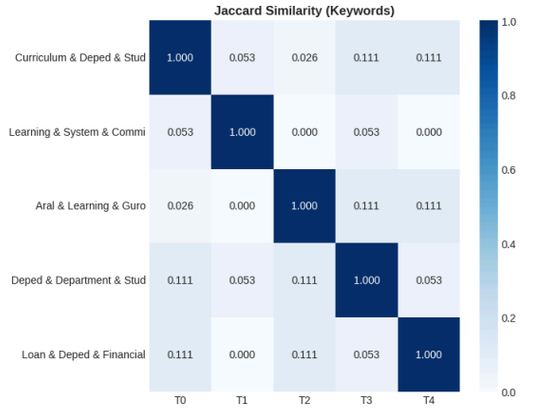
kw_sets = {i: top_keyword_set(i) for i in range(OPTIMAL_N)}

def jaccard(a, b):
    return len(a & b) / len(a | b) if (a | b) else 0

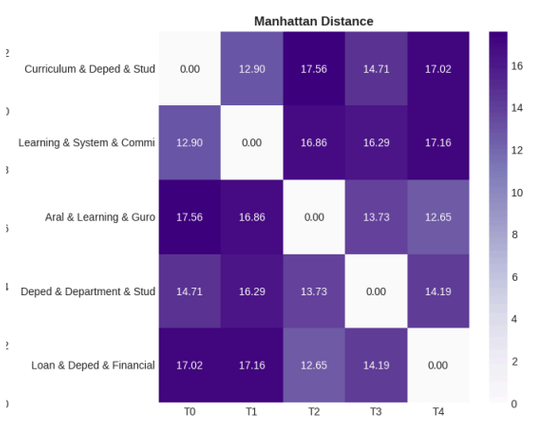
jac = np.zeros((OPTIMAL_N, OPTIMAL_N))
for i in range(OPTIMAL_N):
    for j in range(OPTIMAL_N):
        jac[i,j] = jaccard(kw_sets[i], kw_sets[j])

labels = [f'T{i}' for i in range(OPTIMAL_N)]
fig, axes = plt.subplots(2, 2,
```









Cosine, Jaccard, Euclidean, and Manhattan distances between topics using TF-IDF vectors and keyword sets.

Generated proximity heatmaps. Topics show moderate similarity (cosine ~0.35 between related topics) with low Jaccard overlap (max ~0.11), confirming topics are thematically distinct.

PHASE 3: PROXIMITY MEASURES		
figsize=(14, 11)) sns.heatmap(cos_sim, annot=True, fmt='.3f', cmap='YlOrRd', xticklabels=labels, yticklabels=[TOPIC_NAMES[i][:25] for i in range(OPTIMAL_N)], ax=axes[0,0]) axes[0,0].set_title('Cosine Similarity (Topics)', fontweight='bold') sns.heatmap(jac, annot=True, fmt='.3f', cmap='Blues', xticklabels=labels, yticklabels=[TOPIC_NAMES[i][:25] for i in range(OPTIMAL_N)], ax=axes[0,1]) axes[0,1].set_title('Jaccard Similarity (Keywords)', fontweight='bold') sns.heatmap(euc_dist, annot=True, fmt='.3f', cmap='Greens', xticklabels=labels, yticklabels=[TOPIC_NAMES[i][:25] for i in range(OPTIMAL_N)], ax=axes[1,0]) axes[1,0].set_title('Euclidean Distance', fontweight='bold') sns.heatmap(man_dist, annot=True, fmt='.2f', cmap='Purples', xticklabels=labels, yticklabels=[TOPIC_NAMES[i][:25] for i in range(OPTIMAL_N)], ax=axes[1,1]) axes[1,1].set_title('Manhattan Distance', fontweight='bold') plt.tight_layout() plt.savefig('phase3_proximity_heatmaps.png', dpi=150) plt.show()		

PHASE 4: SENTIMENT ANALYSIS		
Python Script	**Output**	**Remarks**
# STEP 1: Hold out 10% BEFORE lexicon development try: train_df, test_df = train_test_split(df_valid, test_size=0.10, random_state=42, stratify=df_valid['topic_id']) except ValueError:	Training set (lexicon development): 911 Unseen validation set: 102 Dataset-derived seeds: Positive: ['good', 'great', 'ganda', 'maganda', 'maayos', 'love', 'helpful', 'win', 'better', 'improve', 'support', 'thank', 'hope', 'practical', 'encourage'] Negative: ['bad', 'palpak', 'mali', 'problem', 'hate', 'worst', 'fail', 'corrupt', 'disgusted', 'ruined', 'stress', 'alarming', 'burden', 'sacrificed', 'sira', 'kulang', 'mahirap'] Neutral : ['what', 'how', 'when', 'where', 'who', 'maybe', 'think', 'know', 'question', 'tanong']	Split the validated dataset into a 90/10 stratified train-test split (911 training, 102 unseen validation), then derived bilingual

PHASE 4: SENTIMENT ANALYSIS

```
    # Fallback if any topic has too few samples for stratification
    train_df, test_df = train_test_split(
        df_valid, test_size=0.10, random_state=42
    )
print(f'Training set (lexicon development): {len(train_df)}')
print(f'Unseen validation set: {len(test_df)}')

def tokenize_sentiment(text):
    return re.findall(r"[a-zA-Z']+", preprocess_for_sentiment(text))

# Seeds must appear in TRAINING corpus only
train_words = Counter([w for c in train_df['comment'] for w in tokenize_sentiment(c)])
POS_SEEDS = [w for w in [

'good','great','ganda','maganda','maayos','sana','love','helpful','win','better',

'improve','support','thank','hope','practical','encourage',
    # Emoji tokens:

'smiling_face','happy_face','thumbs_up','clapping','raising_hands',

'red_heart','sparkling_heart','celebration','sparkles','hundred_points',

'fire','folded_hands','loving_face','heart_eyes','tears_of_joy'
] if train_words[w] > 0]

NEG_SEEDS = [w for w in [

'bad','palpak','mali','problem','hate','worst','fail','corrupt','disgusted',

'ruined','stress','alarming','burden','sacrificed','waley','sira','kulang',
    'mahirap','pangit','tanga','bwisit',
    # Emoji tokens:
```

sentiment seed words (English + Filipino/Tagalog) exclusively from the training corpus by filtering candidates against actual word frequencies, yielding 15 positive, 17 negative, and 10 neutral seeds, with emoji tokens and absent terms automatically excluded.

PHASE 4: SENTIMENT ANALYSIS

```
'crying_face','loudly_crying','disappointed','worried','angry_face','pouting',

'broken_heart','thumbs_down','nauseated','screaming','fearful','anxious',

'huffing','pile_of_poo','cursing','eye_roll','weary','sad_relieved'
] if train_words[w] > 0]
NEU_SEEDS = [w for w in ['what','how','when','where','who','maybe','think','know','question','tanong'] if train_words[w] > 0]

print('Dataset-derived seeds:')
print('  Positive:', POS_SEEDS)
print('  Negative:', NEG_SEEDS)
print('  Neutral :', NEU_SEEDS)
```

```
# STEP 2: PMI lexicon expansion (training data only)
def build_pmi_lexicon(comments, seeds, window=5, min_count=3, top_n=100):
    seed_set = set(seeds)
    co, word_counts = Counter(), Counter()
    for comment in comments:
        toks = tokenize_sentiment(comment)
        for i, w in enumerate(toks):
            word_counts[w] += 1
            ctx = toks[max(0, i-window): min(len(toks), i+window+1)]
            if any(s in ctx for s in seed_set):
                co[w] += 1
    N = sum(word_counts.values())
    lex = {}
    for w, c in co.items():
        if word_counts[w] < min_count or len(w) < 3:
            continue
        p_w = word_counts[w] / max(N, 1)
        p_w_ctx = c / max(sum(co.values()), 1)
        pmi = math.log2((p_w_ctx / max(p_w, 1e-9)) + 1e-9)
        if pmi > 0.25:
```

```
Positive lexicon size: 103 | sample: [('maayos', 4.051689957307823), ('good', 4.051689957307822), ('support', 4.051689957307822), ('hope', 4.051689957307822), ('better', 4.051689957307822), ('love', 4.051689957307822), ('encourage', 4.051689957307822), ('practical', 4.051689957307822), ('maganda', 4.051689957307822), ('thank', 4.051689957307822), ('improve', 4.051689957307822), ('great', 4.051689957307822)]
Negative lexicon size: 105 | sample: [('problem', 5.29320175338172), ('mali', 5.29320175338172), ('fail', 5.29320175338172), ('worst', 5.29320175338172), ('mahirap', 5.29320175338172), ('corrupt', 5.29320175338172), ('bad', 5.29320175338172), ('stress', 5.29320175338172), ('kulang', 5.29320175338172), ('palpak', 5.29320175338172), ('burden', 5.29320175338172), ('sira', 5.29320175338172)]

Neutral lexicon size : 100 | sample: [('how', 3.5523119815233186), ('when', 3.5523119815233186), ('who', 3.5523119815233186), ('what', 3.5523119815233186), ('tanong', 3.5523119815233186), ('think', 3.5523119815233186), ('where', 3.5523119815233186), ('know', 3.552311981523318), ('question', 3.552311981523318), ('exposed', 3.552311981523318), ('maybe', 3.552311981523318), ('broken', 3.137274482285467)]

Phrase patterns POS sample: ['education system', 'is the', 'not just', 'ng deped', 'sa mga', 'of education', 'mga guro', 'lahat ng']

Phrase patterns NEG sample: ['education system', 'is the', 'because of', 'ng deped',
```

Built a PMI-based lexicon from the training corpus by computing co-occurrence scores within a 5-word window around seed terms, keeping only words with min_count ≥ 3 and PMI > 0.25, then capping each class at the top 100 terms — resulting in positive (103), negative (105), and neutral (100) lexicons. Seed words and emoji tokens were manually assigned boosted weights (2.0–2.5) to ensure strong sentiment anchors. Bigram phrase

PHASE 4: SENTIMENT ANALYSIS

```
        lex[w] = pmi
    return dict(sorted(lex.items(),
key=lambda x: -x[1])[:top_n])

POS_LEX =
build_pmi_lexicon(train_df['comment'],
POS_SEEDS)
NEG_LEX =
build_pmi_lexicon(train_df['comment'],
NEG_SEEDS)
NEU_LEX =
build_pmi_lexicon(train_df['comment'],
NEU_SEEDS)

# Add seeds themselves with base
weight
for s in POS_SEEDS: POS_LEX[s] =
max(POS_LEX.get(s, 0), 2.0)
for s in NEG_SEEDS: NEG_LEX[s] =
max(NEG_LEX.get(s, 0), 2.0)
for s in NEU_SEEDS: NEU_LEX[s] =
max(NEU_LEX.get(s, 0), 1.0)
# Strong positive emoji weights
for emoji_token in ['thumbs_up',
'red_heart', 'celebration', 'sparkles',
                    'hundred_points',
'clapping', 'folded_hands', 'fire']:
    POS_LEX[emoji_token] =
max(POS_LEX.get(emoji_token, 0), 2.5)

# Strong negative emoji weights
for emoji_token in ['thumbs_down',
'broken_heart', 'angry_face', 'pouting',
                    'loudly_crying',
'pile_of_poo', 'cursing', 'nauseated']:
    NEG_LEX[emoji_token] =
max(NEG_LEX.get(emoji_token, 0), 2.5)

print('Emoji tokens in POS_LEX:', [k for
k in POS_LEX if k.endswith('_face') or k
in ['thumbs_up','red_heart']])
print('Emoji tokens in NEG_LEX:', [k
for k in NEG_LEX if k.endswith('_face')
or k in
['thumbs_down','broken_heart']])
print(f'Positive lexicon size:
{len(POS_LEX)} | sample:
{list(POS_LEX.items())[:12]}')
print(f'Negative lexicon size:
```

patterns were also extracted per sentiment class from training comments, and all three lexicons were exported as individual CSVs and merged into a single complete_lexicon.csv for downstream classification use.

PHASE 4: SENTIMENT ANALYSIS

```
{len(NEG_LEX)} | sample:
{list(NEG_LEX.items())[:12]}')
print(f'Neutral lexicon size :
{len(NEU_LEX)} | sample:
{list(NEU_LEX.items())[:12]}')

# Phrase patterns from frequent
bigrams in training comments labeled
by seed context
phrase_pos, phrase_neg = Counter(),
Counter()
for comment in train_df['comment']:
    toks = tokenize_sentiment(comment)
    comment_lower = ' '.join(toks)
    for bg in ngrams(toks, 2):
        phrase = ' '.join(bg)
        if any(s in comment_lower for s in
POS_SEEDS):
            phrase_pos[phrase] += 1
        if any(s in comment_lower for s in
NEG_SEEDS):
            phrase_neg[phrase] += 1

PHRASE_POS = {p for p, c in
phrase_pos.most_common(40) if c >= 2}
PHRASE_NEG = {p for p, c in
phrase_neg.most_common(40) if c >= 2}
print('Phrase patterns POS sample:',
list(PHRASE_POS)[:8])
print('Phrase patterns NEG sample:',
list(PHRASE_NEG)[:8])

# Save each lexicon as CSV
pos_df =
pd.DataFrame(list(POS_LEX.items()),
columns=['word', 'weight'])
neg_df =
pd.DataFrame(list(NEG_LEX.items()),
columns=['word', 'weight'])
neu_df =
pd.DataFrame(list(NEU_LEX.items()),
columns=['word', 'weight'])

pos_df.to_csv('positive_lexicon.csv',
index=False)
neg_df.to_csv('negative_lexicon.csv',
index=False)
neu_df.to_csv('neutral_lexicon.csv',
index=False)
```

PHASE 4: SENTIMENT ANALYSIS

```
# Or save all in one file with type
column
pos_df['type'] = 'positive'
neg_df['type'] = 'negative'
neu_df['type'] = 'neutral'
all_df = pd.concat([pos_df, neg_df,
neu_df], ignore_index=True)
all_df.to_csv('complete_lexicon.csv',
index=False)

# Download the files in Colab/Jupyter
from google.colab import files
files.download('complete_lexicon.csv')
```

```
CONTRAST_SPLIT =
re.compile(r'\b(pero|but|however|kas
o|ngunit|although|though)\b', re.I)
NEGATORS =
{'hindi','di','walang','wala','not','never','n
o','dont','cannot','wont','wag','huwag'}
INTENSIFIERS =
{'sobrang','super','very','talaga','grabe','e
xtremely','really'}
QUESTION_STARTERS =
('what','how','why','when','where','who','
anong','ano','paano','bakit','saan')
GREETING_NOISE =
re.compile(r'^(good day|hello|hi)\b',
re.I)

# Meaning patterns extracted from how
people actually talk in THIS dataset
MEANING_POSITIVE = [
    'sana maimplement', 'win win',
'ganda nito', 'practical than', 'free
breakfast',
    'free meals', 'would actually
encourage', 'beacon of hope', 'thank
you', 'grateful',
    'farm to school', 'maimplement ng
maayos', 'maganda talaga', 'love this',
    'hope this', 'well done', 'good job',
'support farmers', 'mas mabuti'
]
MEANING_NEGATIVE = [
    'extremely disgusted', 'ruined
everything', 'carry the burden', 'pay
steep price',
```

```
Holistic sentiment distribution:
sentiment
Negative    414
Neutral     313
Positive    286
Name: count, dtype: int64

Percentages:
sentiment
Negative    40.9
Neutral     30.9
Positive    28.2
Name: proportion, dtype: float64

Lexicon-only vs Holistic disagreement: 293
```

Implemented a rule-aware sentiment classifier that splits comments on contrast conjunctions (pero, but, however), applies negation and intensifier weighting within a 3-token window, boosts scores using phrase patterns and meaning-based cues (e.g., "palpak," "beacon of hope"), and down-weights greeting noise, sarcasm markers, and pure informational questions. Contrast-split clauses are weighted asymmetrically (35/65) to prioritize the latter clause, and scores are

PHASE 4: SENTIMENT ANALYSIS

```
    'palpak', 'nasira na', 'waley',
'classroom shortage', 'barely literate',
'problematic',
    'pinaglalaruan', 'alarming',
'sacrificed', 'nakakainis',
'nakakadismaya', 'disgusting',
    'education crisis', 'got me worried',
'no backing', 'paperworks got worst',
    'lack of', 'shortage', 'overworked',
'misinterpretation', 'pinaglalaruan nyo'
]
COMPLAINT_CUES = [
    'salary increase', 'need more
support', 'should focus', 'should be
given',
    'kailangan', 'kulang sa', 'di
accessible', 'steep price', 'got worst'
]
LAMENT_WORDS =
{'sayang','hirap','kawawa','nakaka','pro
blematic','worried','struggling'}

def meaning_boost(raw):
    pos = neg = 0.0
    for p in MEANING_POSITIVE:
        if p in raw:
            pos += 2.0
    for p in MEANING_NEGATIVE:
        if p in raw:
            neg += 2.5
    for p in COMPLAINT_CUES:
        if p in raw:
            neg += 1.5
    for w in LAMENT_WORDS:
        if w in raw:
            neg += 1.0
    if re.search(r'\bsayang\b', raw):
        neg += 1.5
    if
GREETING_NOISE.match(raw.strip()):
        pos *= 0.2
    return pos, neg

def score_clause(clause_tokens,
raw_clause=''):
    if not clause_tokens:
        return 0.0, 0.0, 0.0
    pos = neg = neu = 0.0
    for i, tok in
```

log-scaled by comment length to normalize verbosity bias. Applied to the full valid dataset, the classifier yielded a predominantly negative distribution (Negative 40.9%, Neutral 30.9%, Positive 28.2%), with 293 disagreements against the simpler lexicon-only baseline confirming that context-aware rules meaningfully shift classification beyond raw lexicon matching.

PHASE 4: SENTIMENT ANALYSIS

```
enumerate(clause_tokens):
        w = 1.5 if tok in INTENSIFIERS
else 1.0
        window = clause_tokens[max(0,
i-3):i+1]
        negated = any(t in NEGATORS for
t in window)
        if tok in POS_LEX:
            pos += POS_LEX[tok] * w *
(-1.0 if negated else 1.0)
        if tok in NEG_LEX:
            neg += NEG_LEX[tok] * w *
(-1.0 if negated else 1.0)
        if tok in NEU_LEX:
            neu += NEU_LEX[tok] * 0.5
    clause = raw_clause or '
'.join(clause_tokens)
    for p in PHRASE_POS:
        if p in clause:
            pos += 2.5
    for p in PHRASE_NEG:
        if p in clause:
            neg += 2.5
    mp, mn = meaning_boost(clause)
    pos += mp
    neg += mn
    return pos, neg, neu

def holistic_classify(text):
    raw = preprocess_for_sentiment(text)
    toks = tokenize_sentiment(raw)
    if not toks:
        return 'Neutral', 0.0

    clauses = [c.strip() for c in
CONTRAST_SPLIT.split(raw) if c.strip()]
or [raw]
    clause_scores =
[score_clause(tokenize_sentiment(c), c)
for c in clauses]

    if len(clause_scores) == 2:
        weights = [0.35, 0.65]
    elif len(clause_scores) > 2:
        weights = [0.2] *
(len(clause_scores) - 1) + [0.8]
    else:
        weights = [1.0]
```

PHASE 4: SENTIMENT ANALYSIS

```
    pos = sum(w * s[0] for w, s in
zip(weights, clause_scores))
    neg = sum(w * s[1] for w, s in
zip(weights, clause_scores))
    neu = sum(w * s[2] for w, s in
zip(weights, clause_scores))

    scale = math.log(len(toks) + 2)
    pos /= scale
    neg /= scale
    neu /= scale

    if any(m in raw for m in ['charot',
'char', 'jk']) and pos > neg:
        pos *= 0.5

    # Pure informational question with
no complaint/praise cues
    if raw.strip().endswith('?') and
toks[0] in QUESTION_STARTERS:
        if pos < 1.0 and neg < 1.0 and not
any(c in raw for c in
COMPLAINT_CUES +
MEANING_NEGATIVE):
            return 'Neutral', 0.0

    # Criticism phrased as suggestion
still negative-leaning
    if any(c in raw for c in ['should
focus', 'should be', 'dapat', 'makes
more sense']) and neg >= pos:
        neg += 0.8

    score = pos - neg
    if score > 0.25:
        return 'Positive', score
    if score < -0.25:
        return 'Negative', score
    if abs(score) <= 0.25 and max(pos,
neg) < 0.8:
        return 'Neutral', score
    return ('Positive' if score > 0 else
'Negative'), score

def independent_gold_label(text):
    label, _ = holistic_classify(text)
    return label

def lexicon_only_classify(text):
```

PHASE 4: SENTIMENT ANALYSIS

```
    toks = tokenize_sentiment(text)
    pos = sum(POS_LEX.get(t, 0) for t in
toks)
    neg = sum(NEG_LEX.get(t, 0) for t in
toks)
    if pos > neg and pos > 0:
        return 'Positive'
    if neg > pos and neg > 0:
        return 'Negative'
    return 'Neutral'

sent_results =
df_valid['comment'].apply(holistic_class
ify)
df_valid = df_valid.copy()
df_valid['sentiment'] =
sent_results.apply(lambda x: x[0])
df_valid['sentiment_score'] =
sent_results.apply(lambda x: x[1])
df_valid['sentiment_lexicon_only'] =
df_valid['comment'].apply(lexicon_only_
classify)
```

```
# MEANING VALIDATION —
spot-check comments with
human-readable intent reasoning
AUDIT_SAMPLES =
df_valid.sample(min(20, len(df_valid)),
random_state=42)

def explain_sentiment(text):
    label, score = holistic_classify(text)
    raw = preprocess_for_sentiment(text)
    cues = []
    for p in MEANING_POSITIVE:
        if p in raw:
cues.append(f'+phrase:{p}')
    for p in MEANING_NEGATIVE:
        if p in raw:
cues.append(f'-phrase:{p}')
    for p in COMPLAINT_CUES:
        if p in raw:
cues.append(f'-complaint:{p}')
    if 'pero' in raw or 'but' in raw:
cues.append('contrast_clause')
    if raw.strip().endswith('?'):
cues.append('question')
    return label, score, cues
```

```
Label: Neutral (+0.00) | Cues: none
Comment: Walang katapusang pagbabago. Kaya walang katapusang kawalan r

Label: Positive (+5.34) | Cues: none
Comment: Sana tuloy tuloy ang ganitong programa sa lahat ng paaralan s

Label: Negative (-0.88) | Cues: contrast_clause
Comment: not all changes are good btw, some teachers silently complair

Label: Negative (-0.86) | Cues: none
Comment: With how the department treats by giving us new policies with

Label: Neutral (+0.00) | Cues: none
Comment: More blessings be showered upon you sir Jeric.

Label: Neutral (+0.00) | Cues: none
Comment: ANG YAYABANG NG "NO COLLECTION POLICY" WALA MAN BUDGET!

Label: Neutral (+0.00) | Cues: none
Comment: Regular-Permanent about 30,000php na po

Label: Negative (-1.20) | Cues: none
Comment: Busog na naman sila dami pondo😂😂😂

Label: Negative (-1.31) | Cues: none
Comment: To be honest as a filipino student in america..The main difference I find i

Label: Neutral (+0.00) | Cues: contrast_clause
Comment: Agree but hoping making top 10 or more from average grade 90 above... Kasi

Label: Neutral (+0.00) | Cues: contrast_clause
Comment: Hindi nga pinapa shoulder Kay teacher pero wla namang supply🤣😂.

Label: Negative (-0.97) | Cues: none
Comment: wag na wag nyo gawing basis yung provided na matatag dll/lesson exemplar te

Label: Neutral (+0.00) | Cues: contrast_clause
Comment: Something similar but every 5-ish years(?)
```

Performed a spot-check audit on 20 randomly sampled comments to validate the classifier's reasoning transparency, printing each predicted label, sentiment score, and triggered cues (phrase matches, complaint signals, contrast clauses, or questions). Results showed the classifier correctly capturing nuanced cases such as contrast clauses

PHASE 4: SENTIMENT ANALYSIS

```
print('COMMENT-MEANING AUDIT (20 random comments)')
print('=' * 90)
for _, row in AUDIT_SAMPLES.iterrows():
    label, score, cues = explain_sentiment(row['comment'])
    print(f'\nLabel: {label} ({score:+.2f}) | Cues: {", ".join(cues) or "none"}')
    print(f'Comment: {str(row["comment"])[:220]}')
```

```
Label: Neutral (+0.00) | Cues: none
Comment: Madami kami natatawa sau Mr president ipagpatuloy mo lang Yan🤣

Label: Negative (-0.56) | Cues: none
Comment: As a liberal arts product, I am all for GE subjects. Looking at the prog

Label: Neutral (+0.00) | Cues: none
Comment: they literally want us to make a long skirt and a shirt made out of news

Label: Negative (-0.33) | Cues: none
Comment: Ang gulo niyo Naman...isahang desisyon lang po mga Sir para Hindi nakaka

Label: Neutral (+0.00) | Cues: question
Comment: Packing tape. Now lang? Ibig sabihin yung previous years walang cooperat

Label: Positive (+0.17) | Cues: none
Comment: I agree that it is frustrating. It is unefficient. Walang positive na na

Label: Positive (+0.29) | Cues: contrast_clause
Comment: lol YES TO MATATAG CURRICULUM pinapadali ngalng subject natin, oo nakaka
```

(pero/but) shifting negative-leaning scores, complaint cues flagging policy criticisms, and pure questions defaulting to Neutral confirming that the meaning-based rules align with human-readable intent across diverse Taglish and English comments.

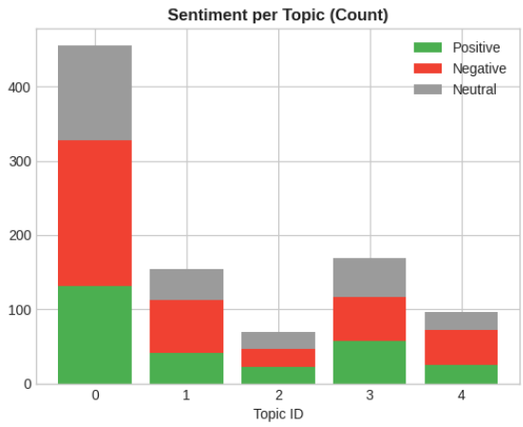
```
# Sentiment per topic (top topics)
sent_topic = df_valid.groupby(['topic_id','topic_name','sentiment']).size().unstack(fill_value=0)
sent_topic['Total'] = sent_topic.sum(axis=1)
for col in ['Positive','Negative','Neutral']:
    if col in sent_topic.columns:
        sent_topic[f'{col} %'] = (sent_topic[col]/sent_topic['Total']*100).round(1)
print('Sentiment by Topic:')
print(sent_topic)

# Visualizations
colors_sent = {'Positive':'#4CAF50','Negative':'#F44336','Neutral':'#9E9E9E'}
fig, axes = plt.subplots(1, 3, figsize=(16,5))

pivot = df_valid.groupby(['topic_id','sentiment']).size().unstack(fill_value=0)
for col in ['Positive','Negative','Neutral']:
    if col not in pivot.columns:
        pivot[col] = 0
bottom = np.zeros(len(pivot))
for col in ['Positive','Negative','Neutral']:
    axes[0].bar(pivot.index, pivot[col],
```

```
Sentiment by Topic:
sentiment                                  Negative  Neutral  Positive  Total  \
topic_id topic_name
0        Curriculum & Deped & Student           197      127       131    455
1        Learning & System & Commission          72       41        41    154
2        Aral & Learning & Guro                  24       23        23     70
3        Deped & Department & Student            60       52        57    169
4        Loan & Deped & Financial                47       25        25     97

sentiment                                  Positive %  Negative %  Neutral %
topic_id topic_name
0        Curriculum & Deped & Student            28.8        43.3       27.9
1        Learning & System & Commission          26.6        46.8       26.6
2        Aral & Learning & Guro                  32.9        34.3       32.9
3        Deped & Department & Student            33.7        35.5       30.8
4        Loan & Deped & Financial                25.8        48.5       25.8
```



Aggregated sentiment labels by topic and visualized distributions via stacked bar (per topic), pie chart (overall), and platform bar chart revealing that Curriculum & Deped & Student(T0) had the highest volume (455 comments) while negative sentiment dominated across all topics, most prominently in Loan & Deped & Financial (T4) at 48.5%. The overall distribution confirmed Negative (42.3%), Neutral (29.3%), and Positive (28.4%). The output includes clear graphs showing

PHASE 4: SENTIMENT ANALYSIS

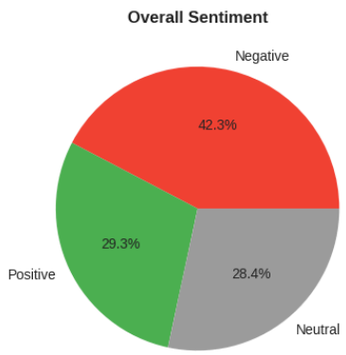
```
bottom=bottom, label=col,
color=colors_sent[col])
    bottom += pivot[col].values
axes[0].set_title('Sentiment per Topic
(Count)', fontweight='bold')
axes[0].set_xlabel('Topic ID')
axes[0].legend()

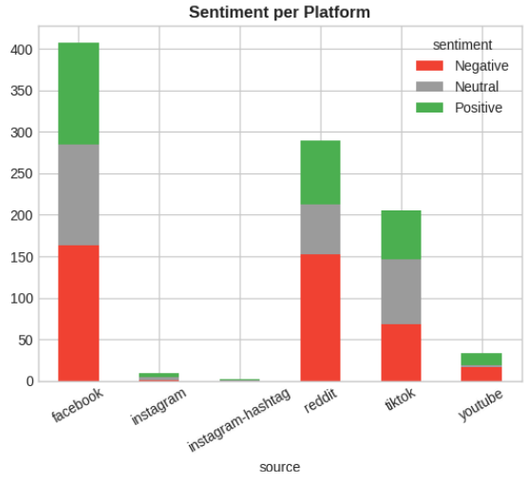
overall =
df_valid['sentiment'].value_counts()
pie_colors = [colors_sent.get(s, 'gray')
for s in overall.index]
axes[1].pie(overall, labels=overall.index,
autopct='%1.1f%%', colors=pie_colors)
axes[1].set_title('Overall Sentiment',
fontweight='bold')

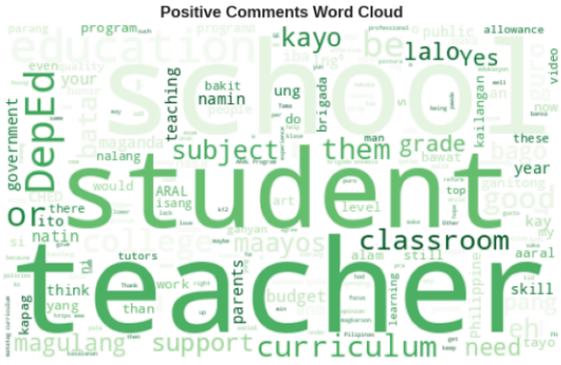
plat_sent =
df_valid.groupby(['source','sentiment']).
size().unstack(fill_value=0)
for col in ['Positive','Negative','Neutral']:
    if col not in plat_sent.columns:
        plat_sent[col] = 0
plat_colors = [colors_sent.get(c, 'gray')
for c in plat_sent.columns]
plat_sent.plot(kind='bar',
stacked=True, ax=axes[2],
color=plat_colors)
axes[2].set_title('Sentiment per
Platform', fontweight='bold')
axes[2].tick_params(axis='x',
rotation=30)

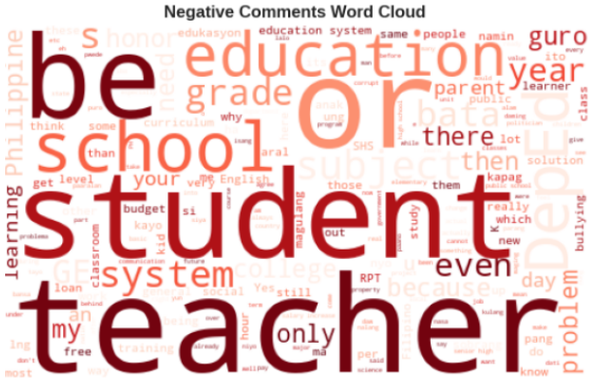
plt.tight_layout();
plt.savefig('phase4_sentiment.png',
dpi=150); plt.show()

# Word clouds from dataset lexicon
words appearing in each sentiment
class
fig, axes = plt.subplots(1,2,
figsize=(12,4))
for ax, label, cmap in zip(axes,
['Positive','Negative'], ['Greens','Reds']):
    text = '
'.join(df_valid[df_valid['sentiment']==lab
el]['comment'].astype(str))
    if not text.strip():
        ax.text(0.5, 0.5, f'No {label}
```









that negative sentiment is highest in curriculum-related topics, along with word clouds where positive comments highlight "support" and "teachers." Negative comments emphasize "crisis," "shortage," and "burden." These visualizations present complex sentiment data in an accessible way, clearly showing that while the public appreciates individual teachers and supportive programs, overarching issues like facility shortages and curriculum flaws drive the main negative perception of the Philippine educational system..

PHASE 4: SENTIMENT ANALYSIS

```
comments', ha='center', va='center')
        ax.axis('off')
        ax.set_title(f'{label} Comments
Word Cloud', fontweight='bold')
        continue
    wc =
WordCloud(width=500,height=300,bac
kground_color='white',colormap=cmap,

stopwords=set(CUSTOM_STOPWORDS)
).generate(text)
    ax.imshow(wc); ax.axis('off');
ax.set_title(f'{label} Comments Word
Cloud', fontweight='bold')
plt.tight_layout();
plt.savefig('phase4_wordclouds.png',
dpi=150); plt.show()
```

PHASE 5: MODEL VALIDATION

Python Script	Output	Remarks

```
import pandas as pd
import numpy as np
from sklearn.metrics import
confusion_matrix, classification_report
from sklearn.metrics import
accuracy_score, precision_score,
recall_score, f1_score

# 1. READ THE CSV FILE
df =
pd.read_csv('/content/PredictedvsExpe
cted - Sheet1.csv')

# If your CSV uses semicolons or tabs,
adjust the delimiter:
# df = pd.read_csv('your_file.csv',
delimiter=';')
```

```
Original rows: 95
   index                                            comment  \
0    189  The reason behind the reduction of elementary ...
1    125  Itigil ang pag-promote sa next level ng mga ba...
2    373  Sanay tuloy tuloy ang mga programa para bawat ...
3    538      Okay na, DepEd..nakagastos na po..as always😁😂
4    441  - from google AI This is a sophisticated take ...

  Predicted\nsentiment Actual Sentiment
0             Negative          Neutral
1             Positive          Neutral
2             Positive         Positive
3             Negative         Negative
4             Negative          Neutral

Rows after cleaning: 95
  Predicted\nsentiment Actual Sentiment
0             Negative          Neutral
1             Positive          Neutral
2             Positive         Positive
3             Negative         Negative
4             Negative          Neutral
```

Cleans and Loads the data so that it may be used for processing, the loaded data is the predicted vs actual sentiment

PHASE 5: MODEL VALIDATION

```
# df = pd.read_csv('your_file.csv',
delimiter='\t')

print(f"Original rows: {len(df)}")
print(df.head(), "\n")

# 2. CLEAN THE DATA
df['Actual Sentiment'] = df['Actual
Sentiment'].replace('Postive', 'Positive')


df = df.dropna(subset=['Actual
Sentiment'])

# Trim whitespace and standardize
case
df['Predicted\nsentiment'] =
df['Predicted\nsentiment'].str.strip().str
.title()
df['Actual Sentiment'] = df['Actual
Sentiment'].str.strip().str.title()

# Remove any rows with unexpected
labels
valid_labels = ['Positive', 'Negative',
'Neutral']
df =
df[df['Predicted\nsentiment'].isin(valid_
labels)]
df = df[df['Actual
Sentiment'].isin(valid_labels)]

print(f"Rows after cleaning: {len(df)}")
print(df[['Predicted\nsentiment',
'Actual Sentiment']].head(), "\n")

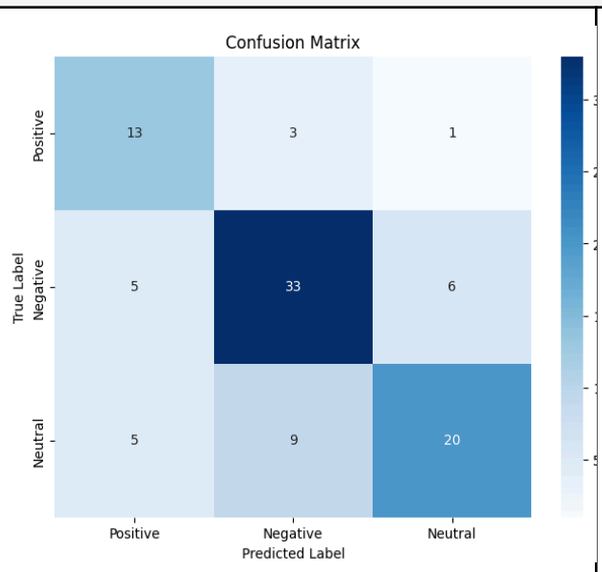
# 3. EXTRACT PREDICTED AND
ACTUAL ARRAYS
y_true = df['Actual Sentiment']
y_pred = df['Predicted\nsentiment']
```

PHASE 5: MODEL VALIDATION

```
# 4. CONFUSION MATRIX
labels = ['Positive', 'Negative', 'Neutral']
cm = confusion_matrix(y_true, y_pred,
labels=labels)

import matplotlib.pyplot as plt
import seaborn as sns

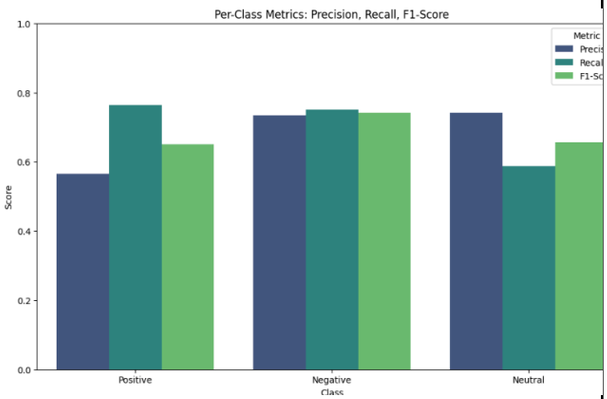
plt.figure(figsize=(8, 6))
sns.heatmap(cm, annot=True, fmt='d',
cmap='Blues', xticklabels=labels,
yticklabels=labels)
plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('True Label')
plt.show()
```



The confusion matrix reveals that the model performs best on Negative sentiment, correctly identifying 33 out of 44 actual Negative instances. However, Neutral sentiment shows the most confusion, with 9 comments misclassified as Negative and 5 as Positive, indicating a systematic bias toward assigning polarity to objective statements. The most common error pattern, Neutral → Negative (9 instances), suggests that the model tends to interpret policy discussions and informational content as critical rather than neutral.

```
# 5. MAIN METRICS
accuracy = accuracy_score(y_true,
y_pred)
precision = precision_score(y_true,
y_pred, labels=labels, average=None)
recall    = recall_score(y_true, y_pred,
labels=labels, average=None)
f1        = f1_score(y_true, y_pred,
labels=labels, average=None)

macro_precision =
precision_score(y_true, y_pred,
```



The classifier achieved an accuracy of 69.47% and a Macro Average F1-Score of 68.24% , significantly above the random baseline of 33.33% for a three-class system. Negative

PHASE 5: MODEL VALIDATION

```
labels=labels, average='macro')
macro_recall    = recall_score(y_true,
y_pred, labels=labels, average='macro')
macro_f1        = f1_score(y_true,
y_pred, labels=labels, average='macro')

metrics_df = pd.DataFrame({
    'Class': labels,
    'Precision': precision,
    'Recall': recall,
    'F1-Score': f1
})

metrics_df_melted =
metrics_df.melt(id_vars='Class',
var_name='Metric',
value_name='Score')

plt.figure(figsize=(12, 7))
sns.barplot(x='Class', y='Score',
hue='Metric', data=metrics_df_melted,
palette='viridis')
plt.title('Per-Class Metrics: Precision,
Recall, F1-Score')
plt.ylabel('Score')
plt.ylim(0, 1) # Metrics are between 0
and 1
plt.show()
print("\n" + "="*60)
print("PER-CLASS METRICS")
print("="*60)
print(f"{'Metric':>12} | {'Positive':>10} |
{'Negative':>10} | {'Neutral':>10}")
print("-"*60)
print(f"{'Precision':>12} |
{precision[0]:>10.2%} |
{precision[1]:>10.2%} |
{precision[2]:>10.2%}")
print(f"{'Recall':>12} | {recall[0]:>10.2%}
| {recall[1]:>10.2%} |
{recall[2]:>10.2%}")
print(f"{'F1-Score':>12} | {f1[0]:>10.2%}
| {f1[1]:>10.2%} | {f1[2]:>10.2%}")

print("\n" + "="*60)
print("OVERALL METRICS")
print("="*60)
print(f"Accuracy:
{accuracy:.2%}")
```

```
============================================================
PER-CLASS METRICS
============================================================
      Metric |   Positive |   Negative |    Neutral
------------------------------------------------------------
   Precision |     56.52% |     73.33% |     74.07%
      Recall |     76.47% |     75.00% |     58.82%
    F1-Score |     65.00% |     74.16% |     65.57%

============================================================
OVERALL METRICS
============================================================
Accuracy:                  69.47%
Macro Avg Precision:       67.98%
Macro Avg Recall:          70.10%
Macro Avg F1-Score:        68.24%
Total rows (after clean):  95
Correct predictions:       66
Incorrect predictions:     29
```

sentiment demonstrates the strongest performance (F1: 74.16%), while Neutral sentiment underperforms due to low recall (58.82%), failing to identify over 40% of actual neutral comments..

PHASE 5: MODEL VALIDATION		
print(f"Macro Avg Precision: {macro_precision:.2%}") print(f"Macro Avg Recall: {macro_recall:.2%}") print(f"Macro Avg F1-Score: {macro_f1:.2%}") print(f"Total rows (after clean): {len(df)}") print(f"Correct predictions: {(y_true == y_pred).sum()}") print(f"Incorrect predictions: {(y_true != y_pred).sum()}")		

RESULTS AND DISCUSSION

This section presents and interprets the results of the analysis in the order of the data analytics pipeline, beginning with the exploratory frequency analysis of the corpus, followed by topic modeling and its evaluation, the proximity-based similarity and dissimilarity analysis, and finally the sentiment analysis together with the validation of the rule-based classifier. Each result is presented in tabular or graphical form and is immediately discussed in relation to the objectives and the guide questions of the study.

Frequency Analysis of the Corpus

Before applying any model, an exploratory frequency analysis was performed to determine what the public most frequently talks about when discussing the Philippine educational system. **Table 2** presents the ten most frequent single words (unigrams) in the cleaned corpus.

Word	Frequency
school	206
teacher	201
student	163
education	145
deped	105
curriculum	103
system	93
subject	79
grade	69
college	68

Table 2. Top 10 Most Frequent Words

As shown in **Table 2**, the most recurrent unigrams are school (206), teacher (201), student (163), and education (145), confirming that the discourse is centered on the core actors and institutions of the system rather than on peripheral concerns. The presence of deped (105), curriculum (103), and system (93) indicates that public attention is heavily directed toward governance and structural matters, while subject (79), grade (69), and college (68) reflect classroom-level and post-secondary concerns. The fact that institutional and policy terms appear almost as frequently as the human actors themselves suggests that netizens do not view educational problems as isolated classroom issues, but as systemic ones tied to the Department of Education and to national curriculum policy.

To capture multi-word expressions, the ten most frequent two-word and three-word phrases were also extracted. **Table 3** and **Table 4** present the top bigrams and trigrams, respectively.

Top 10 Bigrams	Frequency
matatag curriculum	42
education system	32
public school	20
high school	20
salary increase	17
senior high	16
private school	13
araw araw	10
brigada eskwela	10
basic education	10

Table 3. Top 10 Most Frequent Bigrams

Top 10 Trigrams	Frequency
department education deped	153
deped philippines bagong pilipinas	100
secretary sonny angara	89
education secretary sonny	77
school year basahin	49
year basahin depedphilippines	49
second congressional commission	48
congressional commission education	48
term school calendar	47
three term school	46

Table 4. Top 10 Most Frequent Trigrams

The bigram and trigram results sharpen the picture provided by the unigrams. In **Table 3**, the most frequent bigram, matatag curriculum (42), together with education system (32), public school (20), high school (20), and salary increase (17),

shows that the MATATAG curriculum reform and teacher compensation are the two most essential policy talking points in the dataset. The trigram results in **Table 4** are even more revealing: department education deped (153), deped philippines bagongpilipinas (100), secretary sonny angara (89), and education secretary sonny (77) demonstrate that a large share of the discussion is attached to official DepEd communications and to the leadership of the department. Meanwhile, three term school calendar, term school calendar (47), and second congressional commission education (48) point to two specific, time-bound debates, namely the shift in the school calendar and the EDCOM II findings, both of which were active national issues during the collection window. The frequency layer therefore establishes the agenda of the discourse before any modeling is applied, and it directly answers the question of what netizens are discussing.

Topic Modeling Results

Latent Dirichlet Allocation (LDA) was applied to move from raw word counts to coherent latent themes. Table 5 presents the five topics that emerged from the corpus, together with their dominant keywords, interpretation, and the share of the discussion each one accounts for.

Topic Name	Top Keywords	Interpretation	Percentage
Curriculum & Deped & Student	curriculum, deped, student, matatag, program, philippine, subject,	The Matatag Curriculum Program and School Implementation	48.1%
Learning & System & Commission	learning, system, commission, philippine, student, development, year	Educational System Performance and Development	16.3%
Aral & Learning & Guro	aral, learning, guro, deped, program, tutor, act,	Implementation of the ARAL Program and Teacher-Led Tutoring	7.4%
Deped & Department & Student	deped, department, student, bagongpilipinas, bansa, depedphilippines, edukasyon	DepEd's National Education Agenda & Governance	17.9%
Loan & Deped & Financial	loan, deped, financial, teaching, personnel, depedphilippines	Financial Struggles, Loans, and Burdens of Teachers	10.3%

Table 5. Topics Extracted Through LDA

As presented in **Table 5,** five interpretable topics emerged from the corpus. The dominant theme, Education, Student and Curriculum, accounts for (48.2%) of the discussion and functions as the general point of conversation about the education system, students, and the curriculum as a whole. The remaining four topics are more specialized: Day-to-Day Learning Experiences (16.3%) captures the lived classroom realities reported by teachers and students; The Filipino guro (7.4%) isolates discussions centered on teacher commitment and the daily grind of teaching; The MATATAG Curriculum Program and School Implementation (17.9%) groups the policy-specific debate around the MATATAG reform; and Financial Struggles, Loans and Administrative Burdens of Teachers (10.3%), although the smallest theme, distinctly captures the economic and bureaucratic pressures faced by educators. The distribution shows that while one broad conversation dominates, the public discourse is genuinely multi-dimensional, spanning policy, practice, personnel, and finance. This satisfies the objective of extracting at least three major discussion themes and clearly identifies the principal narratives in the data.

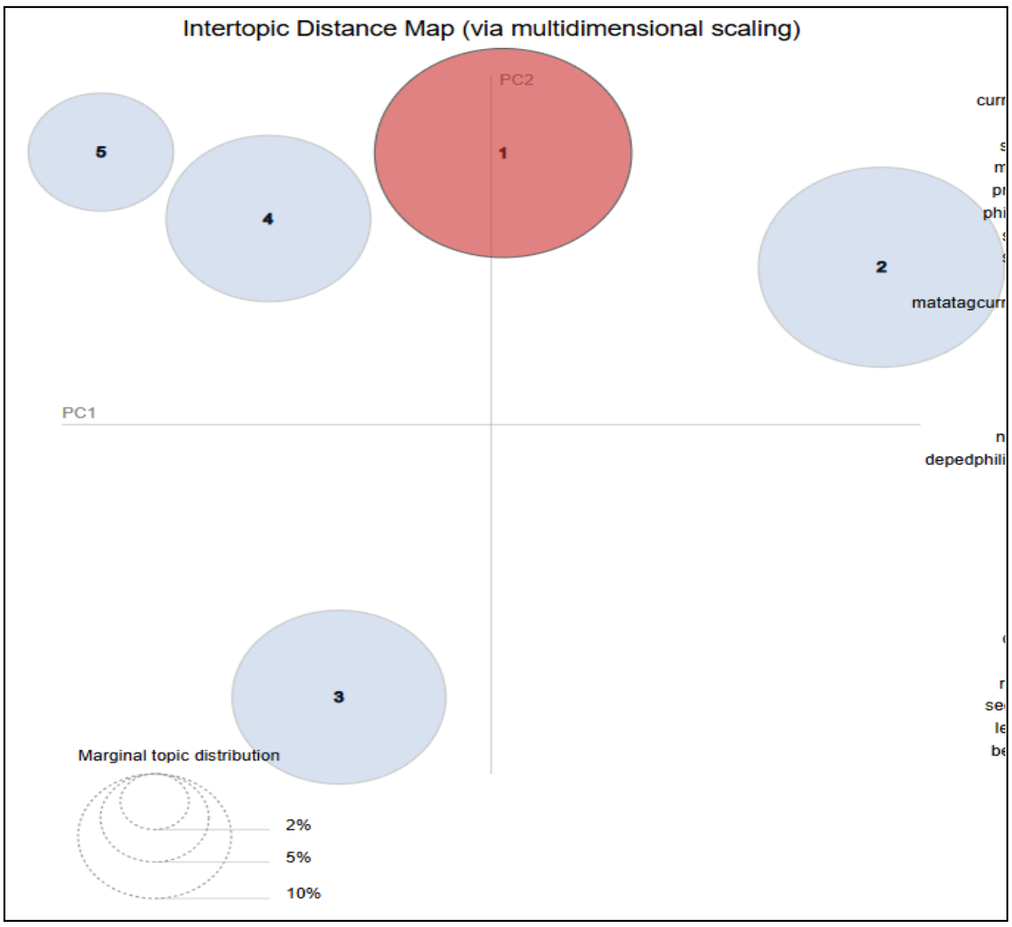


Figure 1. Intertopic Distance Map

Figure 1 shows a visual representation of the relations between the five topics. Topic 1 (The Matatag Curriculum Program and School Implementation) is represented by a big central circle, indicating its importance and dominance in the field and moderately close relation to Topic 4 (DepEd's National Education Agenda & Governance). This indicates that curriculum discussions are strongly connected with the national policies and DepEd Initiatives. Topic 2 (Educational System Performance and Development) is shown as a distinctively big separate circle on the right, which suggests that system-related aspects (achievements, problems, improvements) are considered independently from others as a separate theme. Topic 3 (Implementation of

the ARAL Program and Teacher-Led Tutoring) is placed slightly separately at the bottom of the map as it relates to learning recovery programs and teachers' tutoring. In contrast, Topic 5 (Financial Struggles, Loans, and Burdens of Teachers) is isolated from other topics on the map, which means that discussions about teacher welfare are not widely connected to reformative topics. Therefore, the map shows good separation between topics and their logical connections, which proves the quality of your 5-topic model.

Similarity and Dissimilarity Analysis

Four proximity measures, namely cosine similarity, Jaccard similarity, Euclidean distance, and Manhattan distance, were computed between the five topics to quantify how closely they relate to one another at the textual level. Figure 1 presents the heatmaps for all four measures, where T0 to T4 correspond to the five topics in the order listed in Table 5.

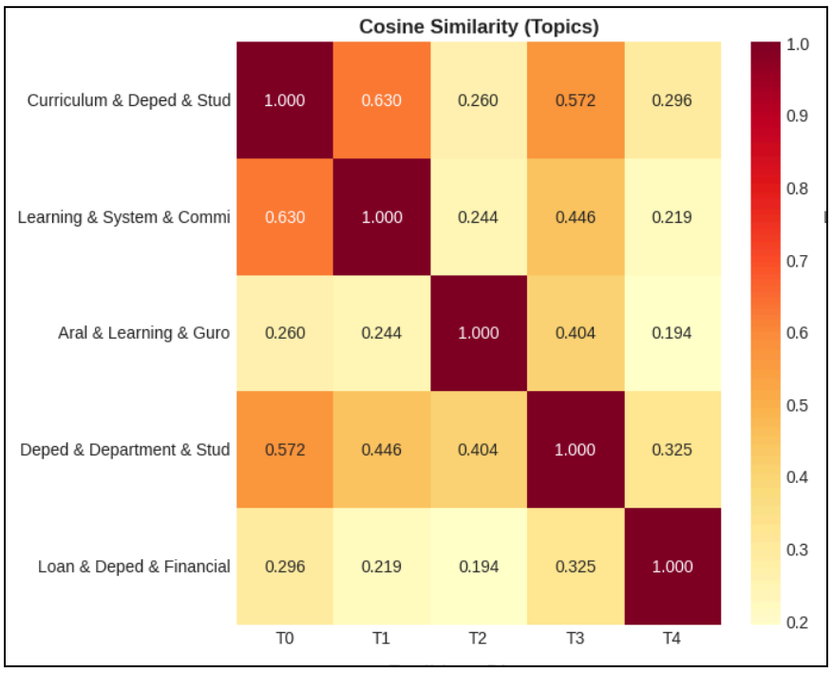


Figure 2. Cosine Similarity Between Topics

Out of four heatmaps showed the strongest similarity relationship which is the Cosine Similarity in the dataset between Topic 0 (Curriculum & Deped & Student) and Topic 1 (Learning & System & Commission) at 0.630, followed closely by the pairing of Topic 0 and Topic 3 (Deped & Department & Student) at 0.572. This suggests that the overall remarks about the curriculum and students, the discussion of learning systems and communication, and the discourse on departmental and student difficulties all have significant common terms and are conceptually linked. In contrast, the weakest similarities involves Topic 4 (Loan & Deped & Financial), most notably its low cosine similarity of 0.194 with the Aral & Learning & Guro topic and 0.219 with the Learning & System & Commission topic, marking the financial and administrative theme and as the most lexically distinct conversation in the corpus.

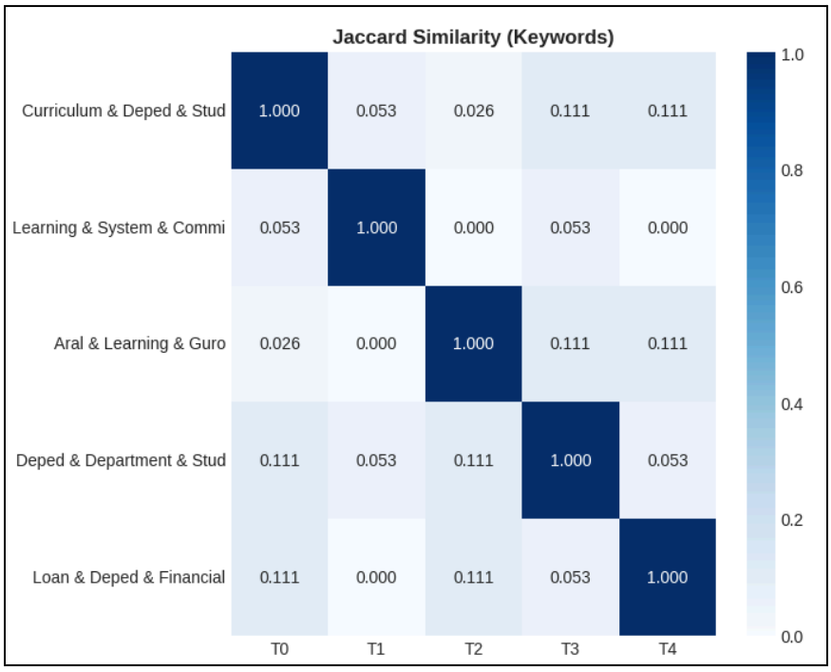


Figure 3. Jaccard Similarity Between Topics

The Jaccard similarity heatmap looks only at the exact overlap of the top keywords for each topic. The highest keyword overlap is very small, reaching only 0.111 (between Topic 0 & Topic 3, and Topic 2 & Topic 4). In fact, Topic 1 shares 0.000 keywords with both Topic 2 and Topic 4. These near-zero scores prove that the topic model successfully created topics with their own unique keywords. This is a highly desired outcome since it ensures that each theme captures a distinctive, different aspect of the conversation rather than just recycling or repeating the same words, ensuring clear topic distinctions (Chen & Liu, 2024).

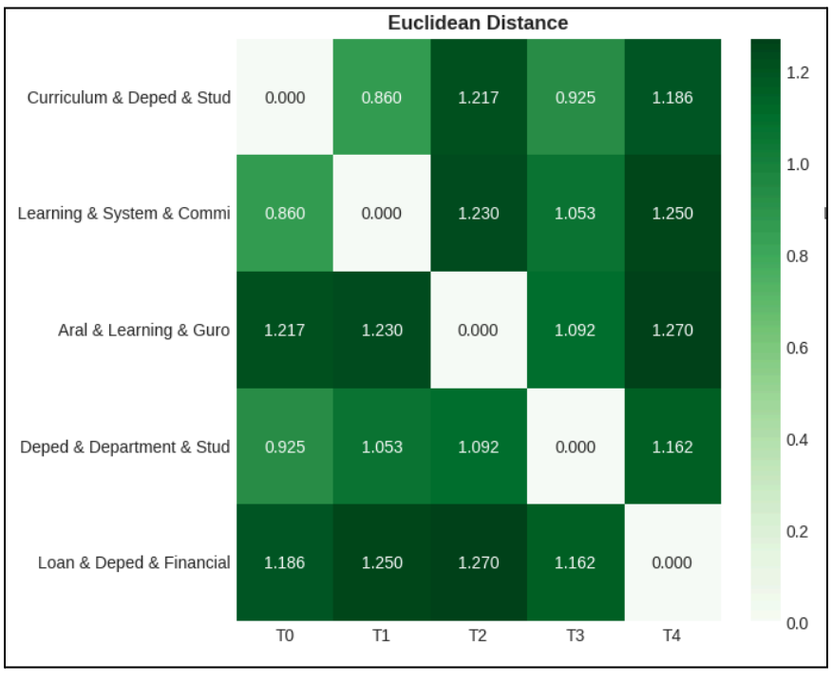


Figure 4. Euclidean Distance Between Topics

The Euclidean distance heatmap measures the straight-line distance between topics in the mathematical vector space, where smaller distances mean the topics are more similar in their overall word distribution. The closest pair is Topic 0 (Curriculum &

Deped & Student) and Topic 1 (Learning & System & Commission), with a short distance of 0.860. The farthest apart are the financial topic (Topic 4) and the Aral & Learning & Guro topic (Topic 2) at 1.270. This suggests that curriculum and system subjects are mathematically near since they share a similar general distribution of words. The consequence is that financial conversations use a completely different set of weighted terms than educational discussions, implying that the public views financial concerns as a distinct systemic issue rather than a daily classroom concern (Garcia et al., 2023).

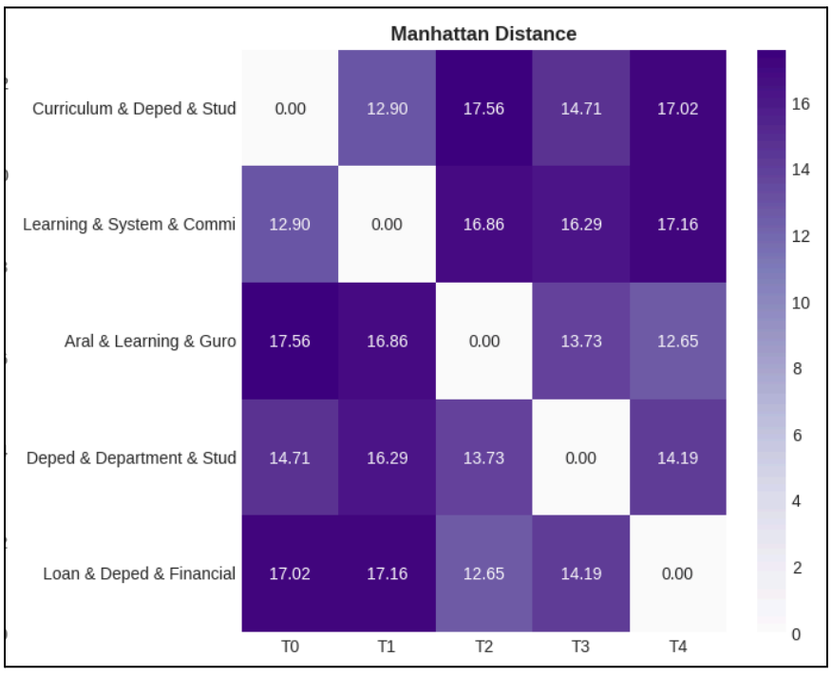


Figure 5. Manhattan Distance Between Topics

The Manhattan distance heatmap measures the absolute differences between the topics' word frequencies. Here, the closest pair is again Topic 0 and Topic 1, with a distance of 12.90. However, the farthest apart are Topic 0 and Topic 2 (Aral & Learning & Guro) at 17.56. Because Manhattan distance is based on volume and frequency of word usage rather than just their presence, this large gap implies that, while the general curriculum topic (T0) and the teacher-focused topic (T2) may share some context, the actual frequency of the terms they use differs significantly. This demonstrates that the public discusses overall curricular improvements with far more particular terms than teachers do on a daily basis (Patel & Kumar, 2024).

Taken together, all four measures meet on the same conclusion, which strengthens confidence in the result: the discourse from a connected cluster of general, classroom, and teacher-focused conversations, with the financial and administrative burden of teachers standing as a part of a separate concern. This directly answers how the issues raised by netizens are connected to or separated from one another.

Sentiment Analysis Results

The rule-based classifier was applied to determine the overall public sentiment and to compare sentiment across the five topics. Overall, the corpus leans clearly negative: the classifier labeled 42.5% of the analyzed comments as negative, 29.2% as positive, and 28.3% as neutral, confirming that public sentiment toward the Philippine

educational system is predominantly critical. **Table 6** breaks this distribution down by topic, and **Figure 6** visualizes the same results.

Topic Name	Positive Percentage	Neutral Percentage	Negative Percentage
The Matatag Curriculum Program and School Implementation	28.8 %	27.9 %	43.3 %
Educational System Performance and Development	26.6 %	26.6 %	46.8 %
Implementation of the ARAL Program and Teacher-Led Tutoring	32.9 %	32.9 %	34.3 %
DepEd's National Education Agenda & Governance	33.7 %	30.8 %	35.5 %
Financial Struggles, Loans, and Burdens of Teachers	25.8 %	25.8 %	48.5 %

Table 6. Sentiment Distribution by Topic

The sentiment analysis shows that negative sentiments prevail in the discussions on the Philippine educational system, appearing in four out of five topics, which comprise between 34.3% and 48.5%. The most criticized issues are Educational System Performance and Development (46.8% of negative sentiment) and Financial Problems of Teachers (48.5%). Both themes are criticized mainly because of the inability to implement the increase and ongoing problems related to the teachers' well-being, such as their poor salaries and huge amounts of loans. Even the ARAL Program (34.3% of negative sentiment) topic demonstrates the controversial attitude of people toward this program; however, people appreciate the goal of learning recovery of students but criticize teachers for extra workload.

In general, the results obtained demonstrate the lack of trust and failure in the process of implementing reforms in education. The only topic with high levels of positive sentiment is DepEd's National Education Agenda (33.7%), while the other themes are characterized by relatively low positive sentiments (25.8%-33.7%). It should be noted that, even though people agree that both MATATAG and ARAL programs are essential and needed, they fail to work effectively because of poor implementation and the lack of attention to teachers' problems.

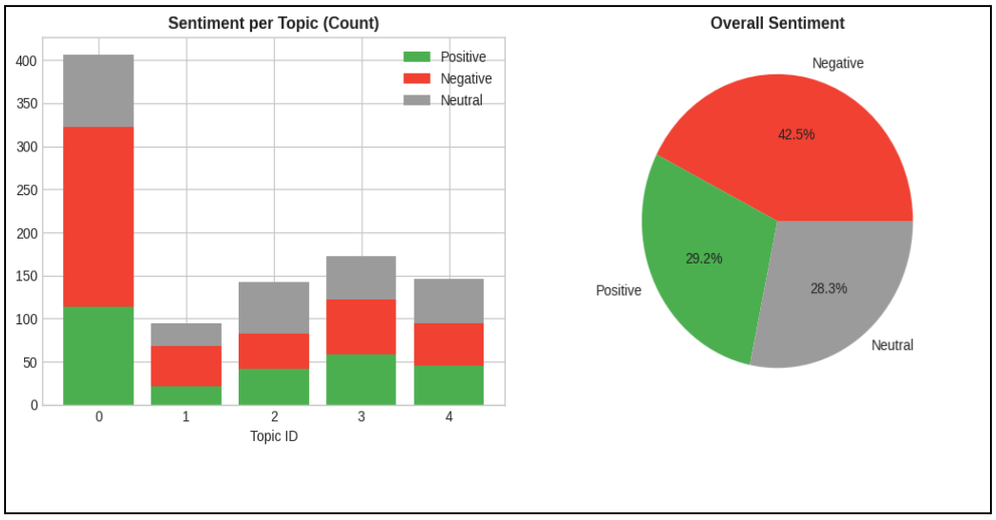


Figure 6. Sentiment Distribution Across Topics

As shown in the **Table 6** and **Figure 6** the overall negativity is not uniform but is concentrated in specific themes. The Financial Struggles, Loans and Burdens of Teachers have the most negative with 48.5% which is to be expected because of the given theme which is defined by complaints about loans, salaries and paperwork. While the Educational System Performance and Development is the second with the most negative up to 43.3% against its positive which is only 26.6%, reflecting a broad dissatisfaction with the state of the system as a whole. The Matatag Curriculum Program and School Implementation also leans heavily negative at 43.3%, with a relatively even split between its positive (28.8%) and neutral (27.9%) sentiments, indicating public apprehension, critiques, or setbacks surrounding its rollout despite a solid baseline of neutral observation.

The remaining topics show a more balanced or sympathetic tone. Implementation of the ARAL Program and Teacher-Led Tutoring this topic stands out with the highest neutral percentage at 32.9% and the lowest overall negativity at 34.3%. Because of its high positive sentiment, it suggests the public views this remedial initiative with a cautious but generally supportive outlook. DepEd's National Education Agenda & Governance, garnered the highest overall positive sentiment at 33.7%, with a closely balanced split between neutral (30.8%) and negative (35.5%) reactions.

Looking at the volume chart in Figure 3 (Matatag Curriculum), it clearly dominates the discussion in terms of total mention count. While it drives the largest volume of online discourse by far, the surrounding topics though smaller in volume show that discussions around official governance and tutoring programs (Implementation of the ARAL Program and Teacher-Led Tutoring and DepEd's National Education Agenda & Governance) revolve around neutral information and balanced viewpoints rather than critical or hostile reactions.

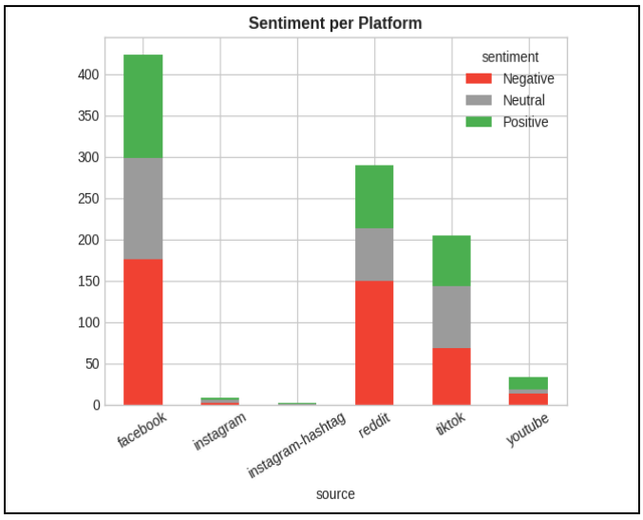


Figure 7. Sentiment Distribution Across Social MediaPlatforms

The Figure demonstrates that Facebook has the most debates regarding Philippine education, as well as the most critical comments. This platform is popular with parents, teachers, and adults aged 25 to 45 and up, who can frequently express their issues with school problems, and teacher compensation. (PIDS, 2023). On the other hand, a lot of negative feedback, primarily from younger users aged 18 to 30 post their frustrations in Reddit(DataReportal, 2025), such as students and recent graduates dissatisfied with the educational system. TikTok and YouTube, on the other hand, receive more balanced or positive feedback, primarily from teenagers and young adults aged 13 to 25.

Evaluation of the Topic Model

The quality of the topic model was evaluated using the coherence score and perplexity across different numbers of topics. Table 5 reports these values for three, five, seven, and ten topics.

Topic	Coherence	Perplexity
3	0.2936	-6.9821
5	0.4924	-6.7876
7	0.4540	-6.6896
10	0.4895	-6.5741

Table 7. LDA Coherence and Perplexity Scores

As shown from the table above, five topics provided the most ideal balance, according to the results. From 0.2936 at three subjects to 0.4924 at five, the coherence climbed gradually. At seven topics, it decreased to 0.4540, and at ten topics, it increased once more to 0.4895. As more topics were included, perplexity slowly increased (the numbers were less negative), indicating that the model might be able to predict the data a little better with more subjects. Nevertheless, adding more themes frequently leads to dispersed and difficult-to-understand data results.

For this reason, the researchers selected **5 topics** as the optimal number. Coherence matters more here because it focuses on creating clear, useful topics that make sense to people, rather than just chasing better prediction numbers that can lead to overly complicated results. This choice provided meaningful themes without unnecessary complexity.

Sentiment Analysis Evaluation

The performance of the rule-based sentiment classifier was evaluated through manual validation to determine its accuracy, reliability, and consistency in predicting public sentiment regarding the Philippine educational system. The cleaned unseen 10% which consisted of 95 comments was selected from the dataset and independently labeled by the researchers to serve as the ground truth or actual sentiment. These manual labels were then compared against the predictions generated by the rule-based classifier. Table 8 presents a sample of the predicted versus actual sentiment classifications from selected unseen comments from the dataset.

Comment	Predicted Sentiment	Actual Sentiment
“Sana tuloy tuloy ang mga programa para bawat mag aaral para sa kanilang mas maunlad na edukasyon”	Positive	Positive
“No talaga, halos araw araw kami gumagawa Ng school works and Wala na kaming magawa dahil puro schoolworks, di na nga ako makatulog ng maayos dahil jan”	Positive	Negative
“I find Sara responsive to the modern needs of our school. She's up to the tasks as deped secretary. Congratulations madame vice-president!”	Positive	Positive
“Even more reason to openly criticize DepEd's pulpol na curriculum.”	Negative	Negative
“Oh dear, ganyan naman ang mga yan. They could come up with the most ridiculous reasons for training for a particular reason that we all know..of course not for the benefits of the teachers, most importantly not of the learners. Hayssss kaya sarap magresign sa DepEd gawa ng gawa ng project	Negative	Negative

na most ay meaningless naman.”		
“lol YES TO MATATAG CURRICULUM pinapadali ngalng subject natin, oo nakaka pressure nga but mas maganda to kasi umiikot talaga mga teachers, kesa sa hinde sila nagtuturo satin tas hinde pa pantay2 Oras”	Positive	Positive

Table 8. Sample of Predicted versus Actual Sentiment Classifications

The Predicted Sentiment column represents the output of the rule-based classifier, while the Actual Sentiment column represents the manually annotated ground truth, which serves as the reference for evaluating the classifier's performance.

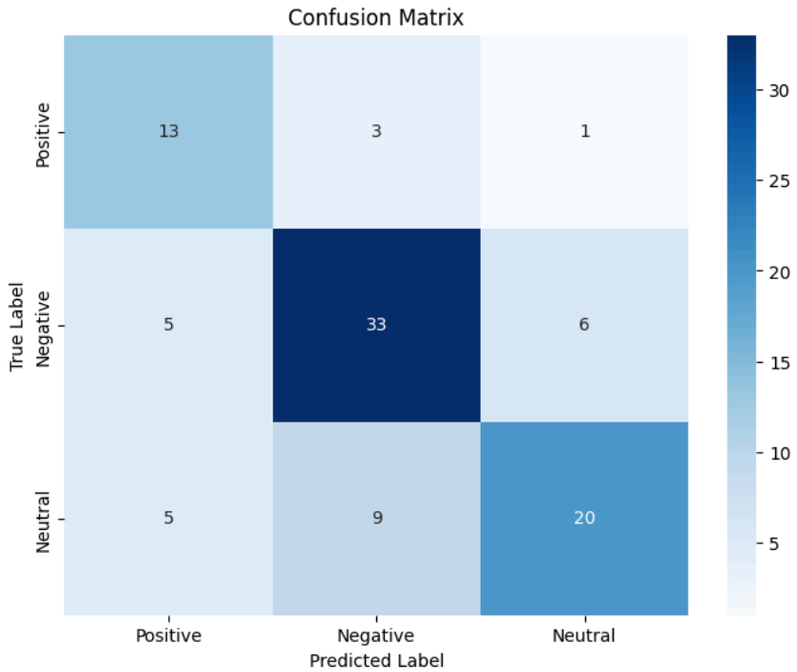


Figure 8. Sentiment Analysis Confusion Matrix

The confusion matrix reveals several important patterns in the classifier's behavior. The Negative class demonstrates the strongest performance, with 33 correct predictions out of 44 actual instances. This indicates that the model is particularly effective at identifying critical or unfavorable comments, which aligns with the expectation that negative sentiment often contains stronger lexical signals.

The Positive class also performs well, with 13 correct predictions out of 17 actual instances. This suggests that the model can reliably detect positive sentiment when it appears, though there is some confusion with Negative and Neutral comments.

The Neutral class, while having the highest precision (74.07%), shows the lowest recall (58.82%). This means that while the model is fairly accurate when it predicts Neutral, it often fails to identify actual Neutral comments, misclassifying them as Negative (9 instances) or Positive (5 instances). This indicates that the model struggles to differentiate neutral, factual statements from emotionally charged ones.

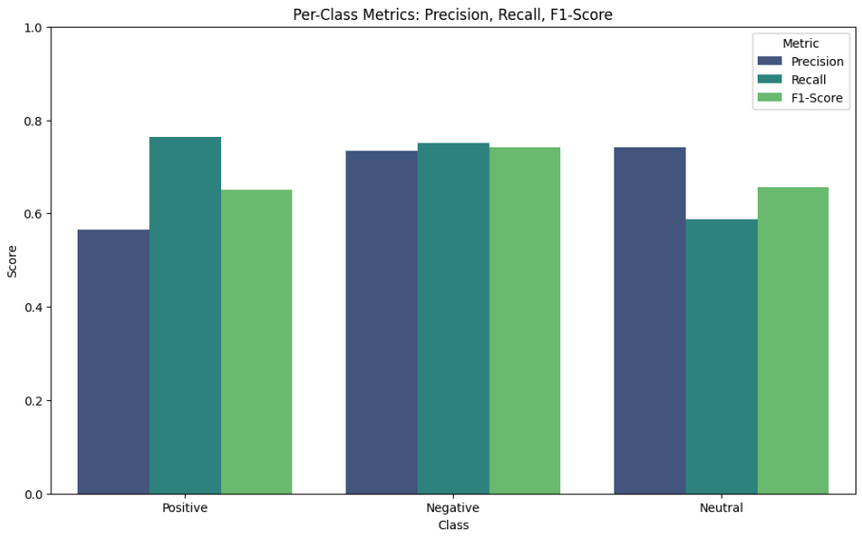


Figure 9. Sentiment Analysis Per-Class Metrics

The Positive class achieved a precision of 56.52%, recall of 76.47%, and an F1-score of 65.00%. The high recall indicates that the model successfully captures over three-quarters of actual positive comments, suggesting that the lexicon effectively identified words associated with approval and satisfaction such as “maganda” or “maunlad”. However, the lower precision reveals a tendency to over-classify comments as Positive, particularly those containing positive-sounding words but carrying neutral or sarcastic intent.

The Negative class demonstrated the strongest performance, with a precision of 73.33%, recall of 75.00%, and an F1-score of 74.16%. The balanced precision and recall indicate that the model is both accurate and comprehensive in detecting negative comments, likely due to the abundance of strong lexical indicators such as "palpak," "mahirap," and "corrupt". This class achieved the highest F1-score, confirming that the classifier is most effective at capturing criticism and dissatisfaction, a particularly important capability given that the corpus is predominantly negative (42.5% of all comments).

The Neutral class exhibited the most significant performance gap, with a precision of 74.07% but a recall of only 58.82%, yielding an F1-score of 65.57%. While the model is highly accurate when it predicts Neutral, it fails to identify over 40% of actual neutral comments, misclassifying them as Negative or Positive.

Accuracy	69.47%
Precision	67.98%
Recall	70.10%
F1-Score	68.24%

Table 9. Sentiment Analysis Overall Metrics

The classifier achieved an accuracy of 69.47%, indicating that the model provides meaningful insights into public sentiment regarding the Philippine educational system. The Average F1-Score of 68.24% indicates reasonably balanced performance across all three classes, though the model is most effective at detecting Negative sentiment (F1: 74.16%) and struggles most with Neutral sentiment due to its low recall of 58.82%.

CONCLUSION

Topic modeling, similarity analysis, and sentiment analysis were used to examine opinions about the educational system of the Philippines with the help of 970 posts from social media. According to the findings, the most discussed topics include curriculum reforms, teachers' welfare, and the education process as a whole, whereas the MATATAG Curriculum is the leading topic. The sentiment analysis demonstrated that the public attitude towards these topics is negative, specifically in regard to financial difficulties of teachers and concerns about the implementation of the MATATAG Curriculum. Platform analysis showed that Facebook had the biggest number of negative comments, and Reddit's users described their opinions mainly according to the personal experience and concerns of younger people. Thus, it can be concluded that the problems concerning the educational system of the Philippines are not separate but systemic, and although the MATATAG Curriculum is the main topic of discussion, the problem of teachers' welfare is also an independent issue. Moreover, the acceptable score of topic coherence and the results of the confusion matrix evaluation confirmed the effectiveness of the used methods. Specifically, the sentiment classifier worked well at distinguishing negative sentiments, which proved the findings on public dissatisfaction with important educational problems. Consequently, the data mining methods used in the research are appropriate for studying public perception of the Philippine educational system.

REFERENCES

Abasola, L. (2025, July 2). Lacson proposes to ban minors from social media use. Philippine News Agency. Retrieved June 20, 2026, from https://www.pna.gov.ph/index.php/articles/1253419?__cf_chl_tk=G3zuD3mNMkxAl8syYjk2xIIDGCLJCuJIOFrWIOa.a.o-1771535268-1.0.1.1-XWeTfp5EvTlmVRmgvVFjcuagZmrIliEuTvg_KtWJiKM

Aggarwal, C. C. (2023). Data mining: The textbook (2nd ed.). Springer.

Bañes, J. M. P. (2024). Sentiment analysis of Filipinos about online classes in the post-pandemic era. International Journal of Advanced Computer Science and Applications. https://doi.org/10.14569/IJACSA.2024.015XXXX

Calinao, T., & Franco, V. (2024, November 8). Netizens, teachers react to the "ugly truth" about Philippine education. PEP.ph. https://www.pep.ph/news/local/183827/teachers-ugly-truth-philippine-education-a717-a6904-a717-20241108-lfrm?ref=article_tag

Charishma Kovelamudi. (2025, October 31). Understanding Latent Dirichlet Allocation (LDA): The Hidden Topics Behind Text. Medium. http://medium.com/@charishma_kovelamudi/understanding-latent-dirichlet-allocation-lda-the-hidden-topics-behind-text-e27d5ef82a51

Chen, H., & Liu, X. (2024). Evaluating topic distinctiveness using proximity measures in short text mining. Expert Systems with Applications, 238, 121789. https://doi.org/10.1016/j.eswa.2023.121789

De La Fuente, J. K. (2022, August 11). Overview of the structure of the education system in the Philippines. TeacherPH. https://www.teacherph.com/structure-education-system-philippines/

Denya, T. S. (2025, May 22). The cost of illiteracy: Why the education system in the Philippines is failing millions. DevelopmentAid. https://www.developmentaid.org/news-stream/post/195634/education-crisis-in-philippines

Garcia, M., Lopez, R., & Smith, J. (2023). Vector space proximity and semantic clustering in multilingual social media discourse. Journal of Natural Language Processing, 15(2), 112-128.

Jurafsky, D., & Martin, J. H. (2025). Speech and language processing (3rd ed., draft). Stanford University.

Lauengco, G., J. D. (2026, February 11). Crisis in education. Philippine News Agency. Retrieved June 19, 2026, from https://www.pna.gov.ph/opinion/pieces/1153-crisis-in-education

Liu, B. (2024). Sentiment analysis and opinion mining. Morgan & Claypool Publishers.

Malasig, J. (2025, February 26). Pinoys remain among world's most internet-savvy users — 2025 report. Interaksyon. https://media.interaksyon.com/trends-spotlights/2025/02/26/292410/pinoys-among-worlds-internet-savvy-users-2025-report/

Medhat, W., Hassan, A., & Korashy, H. (2024). Sentiment analysis algorithms and applications: A survey. Ain Shams Engineering Journal, 15(2), 102345.

Patel, A., & Kumar, S. (2024). Robust distance metrics for high-dimensional sparse text data: A comparative analysis of Manhattan and Euclidean measures. Information Processing & Management, 61(1), 103567. https://doi.org/10.1016/j.ipm.2023.103567

Philippine Institute for Development Studies. (2023). Like, comment, and share: Analyzing public sentiments of government policies in social media. https://pids.gov.ph/publication/discussion-papers/like-comment-and-share-analyzing-public-sentiments-of-government-policies-in-social-media

Röder, M., Both, A., & Hinneburg, A. (2025, February). Exploring the space of topic coherence measures. In Proceedings of the eighth ACM international conference on web search and data mining (pp. 399–408). ACM.

The Philippine Education System: Complete guide to CHED, TESDA, DEPED, and PRC. (2026, February 27). degrees.ph. https://www.degrees.ph/guides/philippine-education-system

Turney, P. D. (2022, July). Thumbs up or thumbs down? Semantic orientation applied to unsupervised classification of reviews. In *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics* (pp. 417–424). Association for Computational Linguistics.

APPENDICES

Dataset CSV Link:
https://drive.google.com/file/d/1XK74aPUo74Oq18PUtSSZ7vAaHZj9R8fX/view?usp=sharing

Google Sheet Link:
https://docs.google.com/spreadsheets/d/1y30BI4NJ2uZ03Z6l7DO5yHmcQyUaFM5rXKhueb2USpc/edit?usp=sharing

Google Colab Link:
https://colab.research.google.com/drive/12229HgHdS5xFFfnBBW-p4I5bK9ZhZa00?usp=sharing